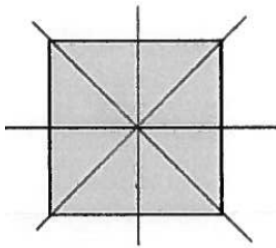
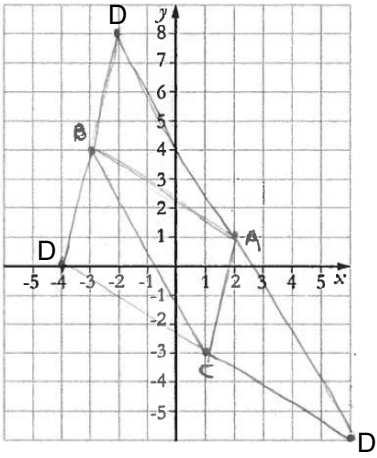


## Bedömningsanvisningar Delprov B

|    |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---|---|---|---|--|--|--|---|--|--|--|---|--|--|--|---|--|--|--|---|--|--|--|
| 1. | $23^\circ$<br>Korrekt svar.                                                                                                                                           | (1/0/0)<br>+E <sub>M</sub>                        | <table border="1"> <tr><td></td><td>E</td><td>C</td><td>A</td></tr> <tr><td>P</td><td></td><td></td><td></td></tr> <tr><td>B</td><td></td><td></td><td></td></tr> <tr><td>M</td><td></td><td></td><td></td></tr> <tr><td>R</td><td></td><td></td><td></td></tr> <tr><td>K</td><td></td><td></td><td></td></tr> </table> |  | E | C | A | P |  |  |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |  |  |
|    | E                                                                                                                                                                     | C                                                 | A                                                                                                                                                                                                                                                                                                                       |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| 2. | <br>En symmetrilinje godtagbart ritad.<br>Samtliga symmetrilinjer godtagbart ritade. | (2/0/0)<br><br>+E <sub>B</sub><br>+E <sub>B</sub> | <table border="1"> <tr><td></td><td>E</td><td>C</td><td>A</td></tr> <tr><td>P</td><td></td><td></td><td></td></tr> <tr><td>B</td><td></td><td></td><td></td></tr> <tr><td>M</td><td></td><td></td><td></td></tr> <tr><td>R</td><td></td><td></td><td></td></tr> <tr><td>K</td><td></td><td></td><td></td></tr> </table> |  | E | C | A | P |  |  |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |  |  |
|    | E                                                                                                                                                                     | C                                                 | A                                                                                                                                                                                                                                                                                                                       |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| 3. | $4,25; 4\frac{1}{4}; \frac{17}{4}$<br>Korrekt svar.                                                                                                                   | (1/0/0)<br>+E <sub>B</sub>                        | <table border="1"> <tr><td></td><td>E</td><td>C</td><td>A</td></tr> <tr><td>P</td><td></td><td></td><td></td></tr> <tr><td>B</td><td></td><td></td><td></td></tr> <tr><td>M</td><td></td><td></td><td></td></tr> <tr><td>R</td><td></td><td></td><td></td></tr> <tr><td>K</td><td></td><td></td><td></td></tr> </table> |  | E | C | A | P |  |  |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |  |  |
|    | E                                                                                                                                                                     | C                                                 | A                                                                                                                                                                                                                                                                                                                       |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| 4. | $\frac{30}{0,97}$<br>Korrekt svar.                                                                                                                                    | (1/0/0)<br>+E <sub>B</sub>                        | <table border="1"> <tr><td></td><td>E</td><td>C</td><td>A</td></tr> <tr><td>P</td><td></td><td></td><td></td></tr> <tr><td>B</td><td></td><td></td><td></td></tr> <tr><td>M</td><td></td><td></td><td></td></tr> <tr><td>R</td><td></td><td></td><td></td></tr> <tr><td>K</td><td></td><td></td><td></td></tr> </table> |  | E | C | A | P |  |  |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |  |  |
|    | E                                                                                                                                                                     | C                                                 | A                                                                                                                                                                                                                                                                                                                       |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| 5. | 17<br>Korrekt svar.                                                                                                                                                   | (1/0/0)<br>+E <sub>B</sub>                        | <table border="1"> <tr><td></td><td>E</td><td>C</td><td>A</td></tr> <tr><td>P</td><td></td><td></td><td></td></tr> <tr><td>B</td><td></td><td></td><td></td></tr> <tr><td>M</td><td></td><td></td><td></td></tr> <tr><td>R</td><td></td><td></td><td></td></tr> <tr><td>K</td><td></td><td></td><td></td></tr> </table> |  | E | C | A | P |  |  |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |  |  |
|    | E                                                                                                                                                                     | C                                                 | A                                                                                                                                                                                                                                                                                                                       |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| 6. | 6<br>Korrekt svar.                                                                                                                                                    | (1/0/0)<br>+E <sub>M</sub>                        | <table border="1"> <tr><td></td><td>E</td><td>C</td><td>A</td></tr> <tr><td>P</td><td></td><td></td><td></td></tr> <tr><td>B</td><td></td><td></td><td></td></tr> <tr><td>M</td><td></td><td></td><td></td></tr> <tr><td>R</td><td></td><td></td><td></td></tr> <tr><td>K</td><td></td><td></td><td></td></tr> </table> |  | E | C | A | P |  |  |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |  |  |
|    | E                                                                                                                                                                     | C                                                 | A                                                                                                                                                                                                                                                                                                                       |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| 7. | $x = 3$<br>Korrekt svar.                                                                                                                                              | (1/0/0)<br>+E <sub>M</sub>                        | <table border="1"> <tr><td></td><td>E</td><td>C</td><td>A</td></tr> <tr><td>P</td><td></td><td></td><td></td></tr> <tr><td>B</td><td></td><td></td><td></td></tr> <tr><td>M</td><td></td><td></td><td></td></tr> <tr><td>R</td><td></td><td></td><td></td></tr> <tr><td>K</td><td></td><td></td><td></td></tr> </table> |  | E | C | A | P |  |  |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |  |  |
|    | E                                                                                                                                                                     | C                                                 | A                                                                                                                                                                                                                                                                                                                       |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| 8. | $v = 60^\circ$<br>Korrekt svar.                                                                                                                                       | (1/0/0)<br>+E <sub>B</sub>                        | <table border="1"> <tr><td></td><td>E</td><td>C</td><td>A</td></tr> <tr><td>P</td><td></td><td></td><td></td></tr> <tr><td>B</td><td></td><td></td><td></td></tr> <tr><td>M</td><td></td><td></td><td></td></tr> <tr><td>R</td><td></td><td></td><td></td></tr> <tr><td>K</td><td></td><td></td><td></td></tr> </table> |  | E | C | A | P |  |  |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |  |  |
|    | E                                                                                                                                                                     | C                                                 | A                                                                                                                                                                                                                                                                                                                       |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K  |                                                                                                                                                                       |                                                   |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |

|        |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---|---|---|---|--|--|--|---|--|--|--|---|--|--|--|---|--|--|--|---|--|--|--|
| 9. a)  | <b>3 €</b><br>Korrekt svar.                                                                                                                                                 | <b>(1/0/0)</b><br>+E <sub>M</sub>                        | <table border="1"> <tr><td></td><td>E</td><td>C</td><td>A</td></tr> <tr><td>P</td><td></td><td></td><td></td></tr> <tr><td>B</td><td></td><td></td><td></td></tr> <tr><td>M</td><td></td><td></td><td></td></tr> <tr><td>R</td><td></td><td></td><td></td></tr> <tr><td>K</td><td></td><td></td><td></td></tr> </table> |  | E | C | A | P |  |  |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |  |  |
|        | E                                                                                                                                                                           | C                                                        | A                                                                                                                                                                                                                                                                                                                       |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| b)     | <b>21 stycken</b><br>Korrekt svar.                                                                                                                                          | <b>(1/0/0)</b><br>+E <sub>M</sub>                        | <table border="1"> <tr><td></td><td>E</td><td>C</td><td>A</td></tr> <tr><td>P</td><td></td><td></td><td></td></tr> <tr><td>B</td><td></td><td></td><td></td></tr> <tr><td>M</td><td></td><td></td><td></td></tr> <tr><td>R</td><td></td><td></td><td></td></tr> <tr><td>K</td><td></td><td></td><td></td></tr> </table> |  | E | C | A | P |  |  |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |  |  |
|        | E                                                                                                                                                                           | C                                                        | A                                                                                                                                                                                                                                                                                                                       |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| c)     | <b>10,50 €; Svar i intervallet 10–11 €</b><br>Godtagbart svar.                                                                                                              | <b>(0/1/0)</b><br>+C <sub>P</sub>                        | <table border="1"> <tr><td></td><td>E</td><td>C</td><td>A</td></tr> <tr><td>P</td><td></td><td></td><td></td></tr> <tr><td>B</td><td></td><td></td><td></td></tr> <tr><td>M</td><td></td><td></td><td></td></tr> <tr><td>R</td><td></td><td></td><td></td></tr> <tr><td>K</td><td></td><td></td><td></td></tr> </table> |  | E | C | A | P |  |  |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |  |  |
|        | E                                                                                                                                                                           | C                                                        | A                                                                                                                                                                                                                                                                                                                       |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| 10.    | Fem tal vars summa är 35 eller fem tal med medianen 9.<br><br>Fem tal vars summa är 35 och medianen är 9, dvs. mittentalet är 9 och två tal $\leq 9$ och två tal $\geq 9$ . | <b>(1/1/0)</b><br>+E <sub>B</sub><br><br>+C <sub>B</sub> | <table border="1"> <tr><td></td><td>E</td><td>C</td><td>A</td></tr> <tr><td>P</td><td></td><td></td><td></td></tr> <tr><td>B</td><td></td><td></td><td></td></tr> <tr><td>M</td><td></td><td></td><td></td></tr> <tr><td>R</td><td></td><td></td><td></td></tr> <tr><td>K</td><td></td><td></td><td></td></tr> </table> |  | E | C | A | P |  |  |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |  |  |
|        | E                                                                                                                                                                           | C                                                        | A                                                                                                                                                                                                                                                                                                                       |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| 11.    | <b>3xy</b><br>Korrekt svar.                                                                                                                                                 | <b>(0/1/0)</b><br>+C <sub>B</sub>                        | <table border="1"> <tr><td></td><td>E</td><td>C</td><td>A</td></tr> <tr><td>P</td><td></td><td></td><td></td></tr> <tr><td>B</td><td></td><td></td><td></td></tr> <tr><td>M</td><td></td><td></td><td></td></tr> <tr><td>R</td><td></td><td></td><td></td></tr> <tr><td>K</td><td></td><td></td><td></td></tr> </table> |  | E | C | A | P |  |  |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |  |  |
|        | E                                                                                                                                                                           | C                                                        | A                                                                                                                                                                                                                                                                                                                       |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| 12.    | Svar större än 0,0025 och mindre än 0,025.<br>Även svar i potensform är korrekt.                                                                                            | <b>(0/2/0)</b><br>+C <sub>B</sub> +C <sub>M</sub>        | <table border="1"> <tr><td></td><td>E</td><td>C</td><td>A</td></tr> <tr><td>P</td><td></td><td></td><td></td></tr> <tr><td>B</td><td></td><td></td><td></td></tr> <tr><td>M</td><td></td><td></td><td></td></tr> <tr><td>R</td><td></td><td></td><td></td></tr> <tr><td>K</td><td></td><td></td><td></td></tr> </table> |  | E | C | A | P |  |  |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |  |  |
|        | E                                                                                                                                                                           | C                                                        | A                                                                                                                                                                                                                                                                                                                       |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| 13. a) | <b>2/5; 4/10; 40 %; 0,4</b><br>Korrekt svar.                                                                                                                                | <b>(1/0/0)</b><br>+E <sub>M</sub>                        | <table border="1"> <tr><td></td><td>E</td><td>C</td><td>A</td></tr> <tr><td>P</td><td></td><td></td><td></td></tr> <tr><td>B</td><td></td><td></td><td></td></tr> <tr><td>M</td><td></td><td></td><td></td></tr> <tr><td>R</td><td></td><td></td><td></td></tr> <tr><td>K</td><td></td><td></td><td></td></tr> </table> |  | E | C | A | P |  |  |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |  |  |
|        | E                                                                                                                                                                           | C                                                        | A                                                                                                                                                                                                                                                                                                                       |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| b)     | <b>1/3; 33 %; 0,33</b><br>Tecknar sannolikheten korrekt för båda hallonbåtarna (6/10 · 5/9).<br>Beräknar sannolikheten och ger ett korrekt svar.                            | <b>(0/1/1)</b><br>+C <sub>M</sub><br>+A <sub>M</sub>     | <table border="1"> <tr><td></td><td>E</td><td>C</td><td>A</td></tr> <tr><td>P</td><td></td><td></td><td></td></tr> <tr><td>B</td><td></td><td></td><td></td></tr> <tr><td>M</td><td></td><td></td><td></td></tr> <tr><td>R</td><td></td><td></td><td></td></tr> <tr><td>K</td><td></td><td></td><td></td></tr> </table> |  | E | C | A | P |  |  |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |  |  |
|        | E                                                                                                                                                                           | C                                                        | A                                                                                                                                                                                                                                                                                                                       |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| 14.    | <b><math>n(n+2); n^2 + 2n</math></b><br>Korrekt svar.                                                                                                                       | <b>(0/0/2)</b><br>+A <sub>P</sub> +A <sub>B</sub>        | <table border="1"> <tr><td></td><td>E</td><td>C</td><td>A</td></tr> <tr><td>P</td><td></td><td></td><td></td></tr> <tr><td>B</td><td></td><td></td><td></td></tr> <tr><td>M</td><td></td><td></td><td></td></tr> <tr><td>R</td><td></td><td></td><td></td></tr> <tr><td>K</td><td></td><td></td><td></td></tr> </table> |  | E | C | A | P |  |  |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |  |  |
|        | E                                                                                                                                                                           | C                                                        | A                                                                                                                                                                                                                                                                                                                       |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K      |                                                                                                                                                                             |                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |

|                      |                                                                                                                                                                                                                                                                                                                                |                                                                                         |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---|---|---|---|--|--|--|---|--|--|--|---|--|--|--|---|--|--|--|---|--|--|--|
| <p><b>15. a)</b></p> | <p>Markerar minst två av punkterna A, B och C korrekt.</p>                                                                                                                                                                                                                                                                     | <p><b>(1/0/0)</b></p> <p>+E<sub>B</sub></p>                                             | <table border="1"> <tr><td></td><td>E</td><td>C</td><td>A</td></tr> <tr><td>P</td><td></td><td></td><td></td></tr> <tr><td>B</td><td></td><td></td><td></td></tr> <tr><td>M</td><td></td><td></td><td></td></tr> <tr><td>R</td><td></td><td></td><td></td></tr> <tr><td>K</td><td></td><td></td><td></td></tr> </table> |  | E | C | A | P |  |  |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |  |  |
|                      | E                                                                                                                                                                                                                                                                                                                              | C                                                                                       | A                                                                                                                                                                                                                                                                                                                       |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P                    |                                                                                                                                                                                                                                                                                                                                |                                                                                         |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B                    |                                                                                                                                                                                                                                                                                                                                |                                                                                         |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M                    |                                                                                                                                                                                                                                                                                                                                |                                                                                         |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R                    |                                                                                                                                                                                                                                                                                                                                |                                                                                         |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K                    |                                                                                                                                                                                                                                                                                                                                |                                                                                         |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| <p><b>b)</b></p>     | <p><b>(-4, 0); (-2, 8) och (6, -6)</b></p> <p>Placerar någon punkt D utifrån punkterna A, B och C så att en parallelogram bildas.</p> <p>Anger koordinaterna för punkten D.</p> <p>Anger koordinaterna för minst två möjliga punkter D.</p>  | <p><b>(0/2/1)</b></p> <p>+C<sub>B</sub></p> <p>+C<sub>M</sub></p> <p>+A<sub>P</sub></p> | <table border="1"> <tr><td></td><td>E</td><td>C</td><td>A</td></tr> <tr><td>P</td><td></td><td></td><td></td></tr> <tr><td>B</td><td></td><td></td><td></td></tr> <tr><td>M</td><td></td><td></td><td></td></tr> <tr><td>R</td><td></td><td></td><td></td></tr> <tr><td>K</td><td></td><td></td><td></td></tr> </table> |  | E | C | A | P |  |  |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |  |  |
|                      | E                                                                                                                                                                                                                                                                                                                              | C                                                                                       | A                                                                                                                                                                                                                                                                                                                       |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P                    |                                                                                                                                                                                                                                                                                                                                |                                                                                         |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B                    |                                                                                                                                                                                                                                                                                                                                |                                                                                         |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M                    |                                                                                                                                                                                                                                                                                                                                |                                                                                         |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R                    |                                                                                                                                                                                                                                                                                                                                |                                                                                         |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K                    |                                                                                                                                                                                                                                                                                                                                |                                                                                         |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |

## Bedömningsanvisningar Delprov C

Bedömningsmatris till uppgift 16, (4/4/4)

| Bedömningen avser                                                                                                                                                                                                                              | Kvalitativa nivåer                                                                                                                                      |                                                                                                                                           |                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                | Lägre                                                                                                                                                   |                                                                                                                                           | Högre                                                                                                                                          |
| <b>Metod och Problemlösning</b><br><br><i>Kvaliteten på de metoder och strategier som eleven använder.</i><br><br><i>Hur väl eleven genomför procedurer och beräkningar.</i><br><br><i>Hur väl eleven tolkar resultat och drar slutsatser.</i> | Bestämmer rätblockets bottenarea korrekt med enhet.<br><br>+E <sub>M</sub>                                                                              | Använder formeln för cirkelns area och bestämmer ett godtagbart värde på $r^2$ eller $(\frac{d}{2})^2$ .<br><br>+C <sub>M</sub>           | Använder formeln för cirkelns area för att bestämma radien genom att använda kvadratroten.<br><br>+A <sub>M</sub>                              |
|                                                                                                                                                                                                                                                | Bestämmer areorna för tårtorna godtagbart, t.ex. genom provning eller har en metod för att bestämma en diameter utifrån en area.<br><br>+E <sub>P</sub> | Löser problemet i c) och bestämmer diametrarna godtagbart för alla tre tårtorna.<br><br>+C <sub>P</sub>                                   | Bestämmer diametrarna för alla tre tårtorna och ger ett rimligt svar med lämplig noggrannhet.<br><br>+A <sub>P</sub>                           |
| <b>Begrepp</b><br><br><i>I vilken grad eleven visar kunskap om matematiska begrepp och samband mellan dessa.</i>                                                                                                                               | Gör korrekt enhetsbyte för att kunna jämföra areor.<br><br>+E <sub>B</sub>                                                                              | Använder förhållandet 1:2:4 (en sjundedel) vid beräkning av tårtornas areor.<br><br>+C <sub>B</sub>                                       | Använder begrepp med stor säkerhet (förhållandet 1:2:4, cirkelns area samt kvadratroten).<br><br>+A <sub>B</sub>                               |
| <b>Kommunikation</b><br><br><i>Kvaliteten på elevens redovisning.</i><br><br><i>Hur väl eleven använder matematiska uttrycksformer (språk och representation).</i>                                                                             | Redovisningen är begriplig och möjlig att följa.<br><br>+E <sub>K</sub>                                                                                 | Redovisningen är lätt att följa och det matematiska språket är acceptabelt och omfattar större delen av uppgiften.<br><br>+C <sub>K</sub> | Redovisningen är välstrukturerad och tydlig med relevant matematiskt språk och terminologi och omfattar hela uppgiften.<br><br>+A <sub>K</sub> |

## Bedömda elevarbeten Delprov C

Bedömda elevarbeten till uppgift 16

Elevarbete 1

a)  $l = 10 \text{ cm}$     $b = 5 \text{ cm}$

$$\begin{array}{r} 10 \\ \times 5 \\ \hline 50 \end{array} \quad (\text{cm})$$

Svar: Bottenarean är  $50 \text{ cm}^2$

b) Nej, det stämmer inte.

$$50 \text{ cm}^2 \times 60 \text{ gäster} = 3000 \text{ cm}^2$$

$$3000 \text{ cm}^2 = 300 \text{ dm}^2$$

Bedömning elevarbete 1

|                | E              | C | A | Poäng | Kommentar |
|----------------|----------------|---|---|-------|-----------|
| Metod          | E <sub>M</sub> |   |   | 1/0/0 |           |
| Problemlösning |                |   |   |       |           |
| Begrepp        |                |   |   |       |           |
| Kommunikation  | E <sub>K</sub> |   |   | 1/0/0 |           |
| Summa          |                |   |   | 2/0/0 |           |

## Elevarbete 2

a)  $10 \cdot 5 = 50$

Tärtbitens bottenarea är  $50 \text{ cm}^2$

b) Varje tärtbit har arean  $50 \text{ cm}^2$

60 gäster

Alltså 60 st tärtbitar

$50 \text{ cm}^2 = 0,5 \text{ dm}^2$  omvandlar  $\text{cm}^2$  till  $\text{dm}^2$

$0,5 \cdot 60 = 30$

tärtan blir  $30 \text{ dm}^2$

c) Diametern

$27 \times 2 = 54$

Bottnvåningen

$27 \times 27 \times \pi = 2289,06$

$13,5 \times 2 = 27$

Mellenvåningen

$13,5 \times 13,5 \times \pi = 572,265$

$6,75 \times 2 = 13,5$

Översta våningen

$6,75 \times 6,75 \times \pi = 143$

$3004,71 \approx 30 \text{ dm}^2$

Om alla gäster ska ha en bit så räcker det här.

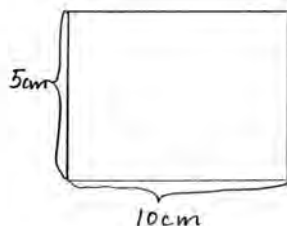
## Bedömning elevarbete 2

|                | E     | C | A | Poäng | Kommentar                                                                |
|----------------|-------|---|---|-------|--------------------------------------------------------------------------|
| Metod          | $E_M$ |   |   | 1/0/0 | Eleven utgår från diameter i stället för area.                           |
| Problemlösning | $E_P$ |   |   | 1/0/0 |                                                                          |
| Begrepp        | $E_B$ |   |   | 1/0/0 | Elevarbetet är möjligt att följa och omfattar endast delar av problemet. |
| Kommunikation  | $E_K$ |   |   | 1/0/0 |                                                                          |
| Summa          |       |   |   | 4/0/0 |                                                                          |

### Elevarbete 3

a) Svar:  $50 \text{ cm}^2$

$$10 \cdot 5 = 50 \text{ cm}^2$$

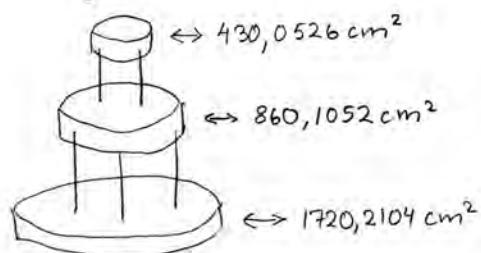


b) Svar:  $30 \text{ dm}^2$

$$50 \cdot 60 = 3000 \text{ cm}^2$$

Sen när jag ska göra om  $\text{cm}^2$  till  $\text{dm}^2$  flyttar jag kommatecknet 2 hakt till vänster och då får jag  $30 \text{ dm}^2$

c) Jag kunde ingen formel för det här så jag testade mig fram och resultatet blev



Det hela tillsammans blir 3010,3683  
Så det står fel på 10.

3000



1256,637061

628,3185307

314,1592654

1963,495408

981,7477042

490,8738521

1720,210473

860,1052367

430,0526183

3010,368328



### Bedömning elevarbete 3

|                | E              | C              | A | Poäng | Kommentar                                                               |
|----------------|----------------|----------------|---|-------|-------------------------------------------------------------------------|
| Metod          | E <sub>M</sub> |                |   | 1/0/0 | Eleven bestämmer de tre tårtornas areor genom att pröva olika diameter. |
| Problemlösning | E <sub>P</sub> |                |   | 1/0/0 |                                                                         |
| Begrepp        | E <sub>B</sub> | C <sub>B</sub> |   | 1/1/0 |                                                                         |
| Kommunikation  | E <sub>K</sub> | C <sub>K</sub> |   | 1/1/0 |                                                                         |
| Summa          |                |                |   | 4/2/0 |                                                                         |

#### Elevarbete 4

a)  $10 \text{ cm} \cdot 5 \text{ cm} = \underline{\underline{50 \text{ cm}^2}}$

b)  $50 \text{ cm}^2 = 0,5 \text{ dm}^2$

$0,5 \text{ dm}^2 \cdot 60 = 30 \text{ dm}^2$

Svar: Ja, det stämmer

c) Totalt  $30 \text{ dm}^2$

Översta tårtan =  $x$

mellersta tårtan =  $2x$

nedersta tårtan =  $4x$

$x + 2x + 4x = 30 \text{ dm}^2$

$7x = 30 \text{ dm}^2$

$x \approx 4,286$

Översta tårtans area  $\approx 4,286 \text{ dm}^2$

$D = 4,286 / \pi \approx 1,36 \text{ dm}$

mellersta tårtans area  $\approx 4,286 \text{ dm}^2 \cdot 2 \approx 8,572 \text{ dm}^2$

$D = 8,572 / \pi \approx 2,73 \text{ dm}$

nedersta tårtans area  $\approx 4,286 \text{ dm}^2 \cdot 4 \approx 17,14 \text{ dm}^2$

$D = 17,14 / \pi \approx 5,46 \text{ dm}$

Svar: Översta tårtans diameter  $\approx 1,36 \text{ dm}$

mellersta tårtans diameter  $\approx 2,73 \text{ dm}$

nedersta tårtans diameter  $\approx 5,46 \text{ dm}$

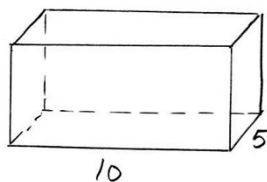
#### Bedömning elevarbete 4

|                | E              | C              | A | Poäng | Kommentar                                                                                                           |
|----------------|----------------|----------------|---|-------|---------------------------------------------------------------------------------------------------------------------|
| Metod          | E <sub>M</sub> |                |   | 1/0/0 | Eleven utgår troligen från formeln för cirkelns omkrets och delar därför bara med $\pi$ för att bestämma diametern. |
| Problemlösning | E <sub>P</sub> |                |   | 1/0/0 |                                                                                                                     |
| Begrepp        | E <sub>B</sub> | C <sub>B</sub> |   | 1/1/0 |                                                                                                                     |
| Kommunikation  | E <sub>K</sub> | C <sub>K</sub> |   | 1/1/0 |                                                                                                                     |
| Summa          |                |                |   | 4/2/0 |                                                                                                                     |



# Elevarbete 5

a) Svar:  $50 \text{ cm}^2$



$$\text{Area} = b \cdot h$$

$$5 \cdot 10 = 50$$

b)  $50 \text{ cm}^2 = 0,5 \text{ dm}^2$

$$\frac{30}{60} = 0,5 \text{ dm}^2 \quad \text{Ja, det stämmer}$$

c)

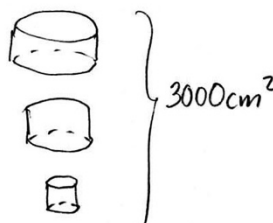
$$\begin{array}{|c|c|c|} \hline x+x+x+x & x+x & x \\ \hline \end{array}$$

3000

$$\text{Area} = \pi \cdot r^2$$

$$x+x+x+x+x+x+x+x = 3000$$

$$\frac{7x}{7} = \frac{3000}{7} = 428,571$$



$$428,571 \cdot 2 = 857,142$$

$$857,142 \cdot 2 = 1714,284$$

$$428,571 + 857,142 + 1714,284 \approx 3000$$

$$1714,284 = 3,14 \cdot r \cdot r$$

$$\frac{1714,284}{3,14} = r \cdot r$$

$$545,950 = r \cdot r$$

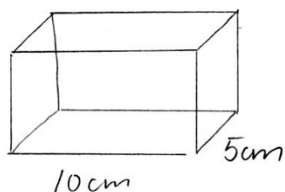
$$272,975 = r$$

## Bedömning elevarbete 5

|                | E              | C              | A | Poäng | Kommentar                                                                                                                                                   |
|----------------|----------------|----------------|---|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Metod          | E <sub>M</sub> | C <sub>M</sub> |   | 1/1/0 | Eleven har delat med 2 i stället för att beräkna kvadratroten. Eleven utgår från formeln för cirkelns area och beräknar $r \cdot r$ korrekt. Gör sedan fel. |
| Problemlösning | E <sub>P</sub> |                |   | 1/0/0 |                                                                                                                                                             |
| Begrepp        | E <sub>B</sub> | C <sub>B</sub> |   | 1/1/0 |                                                                                                                                                             |
| Kommunikation  | E <sub>K</sub> | C <sub>K</sub> |   | 1/1/0 |                                                                                                                                                             |
| Summa          |                |                |   | 4/3/0 |                                                                                                                                                             |

# Elevarbete 6

a)



$$10\text{ cm} \times 5\text{ cm} = 50\text{ cm}^2$$

$$\text{Arean} = 50\text{ cm}^2 \text{ eller } 0,5\text{ dm}^2$$

$$1\text{ dm} \times 0,5\text{ dm} = 0,5\text{ dm}^2$$

b)

60 gäster

$$60 \times 50\text{ cm}^2 = 3000\text{ cm}^2 = 30\text{ dm}^2$$

$$60 \times 0,5\text{ dm}^2 = 30\text{ dm}^2$$

Svar: Ja det stämmer

c)

3000 cm<sup>2</sup>

$$\text{Cirkel} = \text{Arean} = R \cdot R \cdot 3,14$$

$$1750 + 875 + 437,5 = 3062,5$$

$$1:a \text{ cirkeln} = A = 1750\text{ cm}^2 \quad \frac{1750}{3,14} \approx 557\text{ cm}$$

$$23,601 \times 23,601 = 557,007$$

$$23,601 + 23,601 = 47,2$$

$$2:a \text{ cirkeln} = A = \frac{875\text{ cm}^2}{3,14} = 278,6\text{ cm}$$

$$16,7 \times 16,7 = 278,89$$

$$16,7 + 16,7 = 33,4$$

$$3:e \text{ cirkeln} = \frac{437,5}{3,14}\text{ cm}^2 = 139,3$$

$$11,81 \cdot 11,81 = 139,47$$

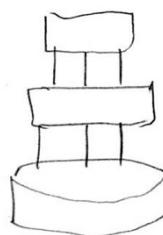
$$11,81 + 11,81 = 23,62$$

Svar

23,62 cm i diameter

33,4 cm i diameter

47,2 cm i diameter



## Bedömning elevarbete 6

|                | E              | C              | A | Poäng | Kommentar                                                                           |
|----------------|----------------|----------------|---|-------|-------------------------------------------------------------------------------------|
| Metod          | E <sub>M</sub> | C <sub>M</sub> |   | 1/1/0 | Eleven gör provningar både vid bestämning av areorna och bestämning av diametrarna. |
| Problemlösning | E <sub>P</sub> | C <sub>P</sub> |   | 1/1/0 | Eleven bestämmer godtagbara diametrar för alla tre tårtorna.                        |
| Begrepp        | E <sub>B</sub> |                |   | 1/0/0 |                                                                                     |
| Kommunikation  | E <sub>K</sub> | C <sub>K</sub> |   | 1/1/0 |                                                                                     |
| Summa          |                |                |   | 4/3/0 |                                                                                     |

# Elevarbete 7

a)  $10 \cdot 5 = 50$

Tårtbitens bottenarea är  $50 \text{ cm}^2$

b)  $30 \text{ dm}^2 = 3000 \text{ cm}^2$

$$\frac{3000 \text{ cm}^2}{60} = 50 \text{ cm}^2$$

Det stämmer. Varje gäst kommer att få sin tårtbit med bottenarean  $50 \text{ cm}^2$ .

c)  $3000 = 7x$

$$x = \frac{3000}{7}$$

$$x \approx 428,6$$

$$\frac{428,6}{3,14} \approx 136,5$$

$$\sqrt{136,5} \approx 11,7$$

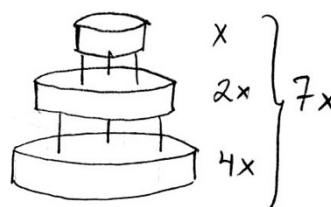
$$11,7 + 11,7 = 23,4$$

Diametern på den:

Översta botten:  $23,4 \text{ cm}$

Mellersta botten:  $23,4 \cdot 2 = 46,8 \text{ cm}$

Understa botten:  $46,8 \cdot 2 = 93,6 \text{ cm}$



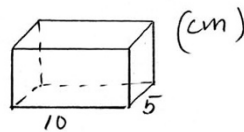
## Bedömning elevarbete 7

|                | E              | C              | A              | Poäng | Kommentar                                                                                                                                                 |
|----------------|----------------|----------------|----------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Metod          | E <sub>M</sub> | C <sub>M</sub> | A <sub>M</sub> | 1/1/1 | Beräknar en diameter korrekt men övergår sedan till längdskala, dvs. använder förhållandet 1:2:4 för diametrarna och får därför inte den andra C-poängen. |
| Problemlösning | E <sub>P</sub> |                |                | 1/0/0 |                                                                                                                                                           |
| Begrepp        | E <sub>B</sub> | C <sub>B</sub> |                | 1/1/0 | Använder förhållandet 1:2:4 vid beräkning av tårtornas area.                                                                                              |
| Kommunikation  | E <sub>K</sub> | C <sub>K</sub> |                | 1/1/0 |                                                                                                                                                           |
| Summa          |                |                |                | 4/3/1 |                                                                                                                                                           |

- a) Jag ska räkna ut bottenarean, alltså basytans area

$$B = \text{längden} \cdot \text{bredden}$$

$$10 \cdot 5 = 50 \quad \text{Svar: } 50 \text{ cm}^2$$



- b) 1 tårtdel =  $50 \text{ cm}^2 = 5 \text{ dm}^2$

$5 \cdot 60 = 300 \text{ dm}^2$  Eftersom det var 60 gäster multiplicerar jag 60 med 5, då får jag reda på hur mycket tårta det kommer gå åt.

Svar: Peters berättelse stämmer inte, det blir  $300 \text{ dm}^2$ , inte  $30 \text{ dm}^2$ .

- c) Hela tårtans area:  $300 \text{ dm}^2$

$$\text{Nedersta: } 171,5 \text{ dm}^2$$

$$\text{Mellersta: } 85,75 \text{ dm}^2$$

$$\text{Översta: } 42,875 \text{ dm}^2$$

$$171,5 + 85,75 + 42,875 = 300 \text{ dm}^2$$

Jag räknade ut hur stor area de olika våningarna skulle ha genom att prova mig fram. Jag tog ett tal tills det stämde och blev  $300 \text{ dm}^2$ .  
ett annat tal



$$r = 7,390 \text{ dm} \quad d = r \cdot 2$$

$$A = 171,5 \text{ dm}^2$$

$$A = \pi \cdot r^2$$

$$3,14 \cdot r^2 = 171,5$$

$$r^2 = \frac{171,5}{3,14}$$

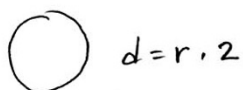
$$r^2 = \pm \sqrt{54,618}$$

$$r = \pm 7,390$$

$$7,390 \cdot 2 = 14,8 \text{ dm}$$

Diametern på den nedersta:

$$14,8 \text{ dm}$$



$$A = 85,75 \text{ dm}^2$$

$$A = \pi \cdot r^2$$

$$3,14 \cdot r^2 = 85,75$$

$$r^2 = \frac{85,75}{3,14}$$

$$r^2 = 27,309$$

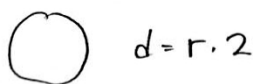
$$r = \pm \sqrt{27,309}$$

$$r = \pm 5,226$$

$$5,226 \cdot 2 = 10,5 \text{ dm}$$

Diametern på den mellersta

$$10,5 \text{ dm}$$



$$A = 42,875$$

$$A = \pi \cdot r^2$$

$$3,14 \cdot r^2 = 42,875$$

$$r^2 = \frac{42,875}{3,14}$$

$$r^2 = \pm \sqrt{13,654}$$

$$r = \pm 3,695$$

$$3,695 \cdot 2 = 7,4 \text{ dm}$$

Diametern på den översta :

$$7,4 \text{ dm}$$

#### Bedömning elevarbete 8

|                | E              | C              | A              | Poäng | Kommentar                                                                                               |
|----------------|----------------|----------------|----------------|-------|---------------------------------------------------------------------------------------------------------|
| Metod          | E <sub>M</sub> | C <sub>M</sub> | A <sub>M</sub> | 1/1/1 | Gör alla beräkningar utifrån 300 dm <sup>2</sup> (följdfel).                                            |
| Problemlösning | E <sub>P</sub> | C <sub>P</sub> |                | 1/1/0 | Detta fel påverkar inte svårigheten i de fortsatta beräkningarna men ger orimligt svar.                 |
| Begrepp        |                |                |                | 0/0/0 | Eleven gör fel vid enhetsbytet i b) och använder ingen generell metod vid beräkning av tårtornas areor. |
| Kommunikation  | E <sub>K</sub> | C <sub>K</sub> | A <sub>K</sub> | 1/1/1 |                                                                                                         |
| Summa          |                |                |                | 3/3/2 |                                                                                                         |

# Elevarbete 9

a)  $50 \text{ cm}^2$

$$10 \cdot 5 = 50 \text{ cm}^2$$

Eftersom längden och bredden finns är det bara att multiplicera dem så får man arean på bottenplattan.

b) Peter räknade fel

Svar:  $300 \text{ dm}^2$

$$30 \text{ dm}^2 = 300 \text{ cm}^2$$

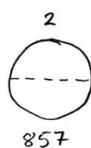
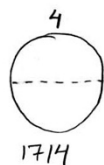
$$50 \cdot 60 = 3000 \text{ cm}^2$$

$$300 \text{ dm}^2 = 3000 \text{ cm}^2$$

Bitarna är  $50 \text{ cm}^2$  styck och det är 60 pers. som kommer alltså måste man multiplicera 50 och 60 med varandra.

Peter multiplicerade  $60 \cdot 50$  men glömde en nolla i svaret.

c) Stora Mellan Lilla  
47cm 33cm 23cm



$$\frac{3000}{7} \approx 428,5 \text{ (cm}^2\text{)}$$

$$\frac{1714}{\pi} \approx 545,5$$

$$\frac{428,5}{\pi} \approx 136,3$$

$$\sqrt{545,5} \approx 23,3$$

$$\sqrt{136,3} \approx 11,6$$

$$23,3 \cdot 2 = 46,6 \text{ cm} \approx 47 \text{ cm}$$

$$11,6 \cdot 2 = 23,2$$

$$\frac{857}{\pi} \approx 272,7$$

$$\sqrt{272,7} \approx 16,5$$

$$16,5 \cdot 2 = 33$$

## Bedömning elevarbete 9

|                | E              | C              | A              | Poäng | Kommentar                                               |
|----------------|----------------|----------------|----------------|-------|---------------------------------------------------------|
| Metod          | E <sub>M</sub> | C <sub>M</sub> | A <sub>M</sub> | 1/1/1 |                                                         |
| Problemlösning | E <sub>P</sub> | C <sub>P</sub> | A <sub>P</sub> | 1/1/1 |                                                         |
| Begrepp        |                | C <sub>B</sub> | A <sub>B</sub> | 0/1/1 | Gör fel i enhetsbytet i b) och får därför inte E-poäng. |
| Kommunikation  | E <sub>K</sub> | C <sub>K</sub> |                | 1/1/0 |                                                         |
| Summa          |                |                |                | 3/4/3 |                                                         |

Elevarbete 10

a)  $10 \cdot 5 = 50$  Svar: Bottenarean är  $50 \text{ cm}^2$ .

b)  $60 \cdot 50 = 3000 \text{ cm}^2 = 30 \text{ dm}^2$

Svar: Ja, det stämmer

60 st gäster ska få en tårtbit på  $50 \text{ cm}^2$ , räknar man ihop allas tårtbitar ( $60 \cdot 50$ ) blir det en arean på  $3000 \text{ cm}^2$ . Med  $100 \text{ cm}^2 = 1 \text{ dm}^2$  blir då  $3000 \text{ cm}^2 = 30 \text{ dm}^2$

c) Översta våningen =  $x$

Andra våningen =  $2x$

Första våningen =  $4x$

$$x + 2x + 4x = 3000 \text{ cm}^2$$

$$7x = 3000$$

$$\frac{7x}{7} = \frac{3000}{7}$$

$$x \approx 430$$

$$\text{Cirkeln area} = r^2 \cdot \pi$$

$$\frac{430}{\pi} \approx 136,87 \quad \sqrt{136,87} \approx 11,70 \approx 12 \quad \left. \vphantom{\frac{430}{\pi}} \right\} 24$$

$$\frac{860}{\pi} \approx 273,75 \quad \sqrt{273,75} \approx 16,55 \approx 17 \quad \left. \vphantom{\frac{860}{\pi}} \right\} 34$$

$$\frac{1720}{\pi} \approx 547,49 \quad \sqrt{547,49} \approx 23,40 \approx 24 \quad \left. \vphantom{\frac{1720}{\pi}} \right\} 48$$

Först räknade jag ut hur mycket area var och en tårtbit ska ha. Avrunda uppåt för det är lättare att tårtan blir för stor än för liten då kanske det inte räcker till alla. Sen delade jag varje area med  $\pi$  för att sedan ta roten ur så jag fick veta radien. Sedan är 2 radien = diametern.

Svar: 24 cm, 34 cm, 48 cm är diametrarna.

Bedömning elevarbete 10

|                | E              | C              | A              | Poäng | Kommentar |
|----------------|----------------|----------------|----------------|-------|-----------|
| Metod          | E <sub>M</sub> | C <sub>M</sub> | A <sub>M</sub> | 1/1/1 |           |
| Problemlösning | E <sub>P</sub> | C <sub>P</sub> | A <sub>P</sub> | 1/1/1 |           |
| Begrepp        | E <sub>B</sub> | C <sub>B</sub> | A <sub>B</sub> | 1/1/1 |           |
| Kommunikation  | E <sub>K</sub> | C <sub>K</sub> | A <sub>K</sub> | 1/1/1 |           |
| Summa          |                |                |                | 4/4/4 |           |

## Bedömningsanvisningar Delprov D

|     |                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                         |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---|---|---|---|--|--|--|---|--|--|--|---|--|--|--|---|--|--|--|---|--|--|--|
| 17. | <p><b>34 personer; 34</b></p> <p>Påbörjad lösning, t.ex. beräknar antalet som äter lasagne och/eller andelen som äter fiskgratäng.</p> <p>Redovisad godtagbar metod vid beräkning av antal eller andel.</p> <p>Redovisning med korrekt svar.</p>                                                                                                                                           | <p><b>(3/0/0)</b></p> <p>+E<sub>P</sub></p> <p>+E<sub>M</sub></p> <p>+E<sub>K</sub></p> | <table border="1"> <tr><td></td><td>E</td><td>C</td><td>A</td></tr> <tr><td>P</td><td></td><td></td><td></td></tr> <tr><td>B</td><td></td><td></td><td></td></tr> <tr><td>M</td><td></td><td></td><td></td></tr> <tr><td>R</td><td></td><td></td><td></td></tr> <tr><td>K</td><td></td><td></td><td></td></tr> </table> |  | E | C | A | P |  |  |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |  |  |
|     | E                                                                                                                                                                                                                                                                                                                                                                                          | C                                                                                       | A                                                                                                                                                                                                                                                                                                                       |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P   |                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                         |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B   |                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                         |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M   |                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                         |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R   |                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                         |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K   |                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                         |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| 18. | <p><b>80 apelsiner, 128 bananer och 192 äpplen</b></p> <p>Beräknar andelar eller beräknar hur många gånger större den nya helheten är.</p> <p>Löser problemet och ger korrekt svar.</p> <p>Redovisningen är möjlig att följa, delberäkningar visas.</p> <p> Till uppgiften finns bedömda elevarbeten.</p> | <p><b>(3/0/0)</b></p> <p>+E<sub>P</sub></p> <p>+E<sub>M</sub></p> <p>+E<sub>K</sub></p> | <table border="1"> <tr><td></td><td>E</td><td>C</td><td>A</td></tr> <tr><td>P</td><td></td><td></td><td></td></tr> <tr><td>B</td><td></td><td></td><td></td></tr> <tr><td>M</td><td></td><td></td><td></td></tr> <tr><td>R</td><td></td><td></td><td></td></tr> <tr><td>K</td><td></td><td></td><td></td></tr> </table> |  | E | C | A | P |  |  |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |  |  |
|     | E                                                                                                                                                                                                                                                                                                                                                                                          | C                                                                                       | A                                                                                                                                                                                                                                                                                                                       |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P   |                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                         |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B   |                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                         |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M   |                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                         |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R   |                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                         |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K   |                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                         |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| 19. | <p><b>42 bröd; 42</b></p> <p>Påbörjad lösning, t.ex. beräknar skillnad i pris för hembakat respektive köpt bröd.</p> <p>Redovisad godtagbar metod vid beräkning av antalet bröd.</p> <p>Tolkar resultatet och ger korrekt svar.</p> <p> Till uppgiften finns bedömda elevarbeten.</p>                   | <p><b>(2/1/0)</b></p> <p>+E<sub>P</sub></p> <p>+E<sub>M</sub></p> <p>+C<sub>P</sub></p> | <table border="1"> <tr><td></td><td>E</td><td>C</td><td>A</td></tr> <tr><td>P</td><td></td><td></td><td></td></tr> <tr><td>B</td><td></td><td></td><td></td></tr> <tr><td>M</td><td></td><td></td><td></td></tr> <tr><td>R</td><td></td><td></td><td></td></tr> <tr><td>K</td><td></td><td></td><td></td></tr> </table> |  | E | C | A | P |  |  |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |  |  |
|     | E                                                                                                                                                                                                                                                                                                                                                                                          | C                                                                                       | A                                                                                                                                                                                                                                                                                                                       |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P   |                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                         |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B   |                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                         |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M   |                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                         |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R   |                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                         |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K   |                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                         |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| 20. | <p><b>16 kronor</b></p> <p>Påbörjad lösning, t.ex. beräknar vad ett glas juice kostar.</p> <p>Löser problemet och ger korrekt svar.</p> <p>Tydlig redovisning med lämpligt matematiskt språk.</p> <p> Till uppgiften finns bedömda elevarbeten.</p>                                                     | <p><b>(1/2/0)</b></p> <p>+E<sub>P</sub></p> <p>+C<sub>P</sub></p> <p>+C<sub>K</sub></p> | <table border="1"> <tr><td></td><td>E</td><td>C</td><td>A</td></tr> <tr><td>P</td><td></td><td></td><td></td></tr> <tr><td>B</td><td></td><td></td><td></td></tr> <tr><td>M</td><td></td><td></td><td></td></tr> <tr><td>R</td><td></td><td></td><td></td></tr> <tr><td>K</td><td></td><td></td><td></td></tr> </table> |  | E | C | A | P |  |  |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |  |  |
|     | E                                                                                                                                                                                                                                                                                                                                                                                          | C                                                                                       | A                                                                                                                                                                                                                                                                                                                       |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P   |                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                         |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B   |                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                         |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M   |                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                         |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R   |                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                         |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K   |                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                         |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |



|        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---|---|---|---|--|--|--|---|--|--|--|---|--|--|--|---|--|--|--|---|--|--|--|
| 21. a) | Svar i intervallet 30–40 %.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | (1/0/0)<br>+E <sub>B</sub>                                               | <table border="1"> <tr><td></td><td>E</td><td>C</td><td>A</td></tr> <tr><td>P</td><td></td><td></td><td></td></tr> <tr><td>B</td><td></td><td></td><td></td></tr> <tr><td>M</td><td></td><td></td><td></td></tr> <tr><td>R</td><td></td><td></td><td></td></tr> <tr><td>K</td><td></td><td></td><td></td></tr> </table> |  | E | C | A | P |  |  |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |  |  |
|        | E                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | C                                                                        | A                                                                                                                                                                                                                                                                                                                       |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| b)     | <p><b>Enkel förklaring som bygger på att det är olika länder med olika mycket hushållsavfall, som ingår i diagrammen, t.ex.</b></p> <p>”Därför att Island har mindre hushållsavfall än Danmark.”</p> <p><b>Någon förklaring där det framgår att helheterna är olika. Formulering som bygger på resonemang om Sveriges andel i relation till övriga länder, t.ex.</b></p> <p>”Helheten är olika eftersom olika länder ingår. I det första diagrammet finns inte Danmark med och i det andra inte Island. Eftersom Danmark står för nästan lika mycket avfall som Sverige, och Island står för bara en väldigt liten del jämfört med Sverige, blir Sveriges andel olika stor i diagrammen.”</p> | (1/1/0)<br><br>+E <sub>R</sub><br><br><br>+C <sub>R</sub>                | <table border="1"> <tr><td></td><td>E</td><td>C</td><td>A</td></tr> <tr><td>P</td><td></td><td></td><td></td></tr> <tr><td>B</td><td></td><td></td><td></td></tr> <tr><td>M</td><td></td><td></td><td></td></tr> <tr><td>R</td><td></td><td></td><td></td></tr> <tr><td>K</td><td></td><td></td><td></td></tr> </table> |  | E | C | A | P |  |  |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |  |  |
|        | E                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | C                                                                        | A                                                                                                                                                                                                                                                                                                                       |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| 22.    | <p><b>36 salamikorvar; 36</b></p> <p>Påbörjad lösning, t.ex. visar hur stor del av antalet korvar som är sålda.</p> <p>Lösning som visar andelar, t.ex. att 5/12 motsvarar 15 korvar eller korrekta proportioner.</p> <p>Löser problemet och ger korrekt svar.</p> <p> Till uppgiften finns bedömda elevarbeten.</p>                                                                                                                                                                                                                                                                                       | (1/2/0)<br><br>+E <sub>B</sub><br><br>+C <sub>M</sub><br>+C <sub>P</sub> | <table border="1"> <tr><td></td><td>E</td><td>C</td><td>A</td></tr> <tr><td>P</td><td></td><td></td><td></td></tr> <tr><td>B</td><td></td><td></td><td></td></tr> <tr><td>M</td><td></td><td></td><td></td></tr> <tr><td>R</td><td></td><td></td><td></td></tr> <tr><td>K</td><td></td><td></td><td></td></tr> </table> |  | E | C | A | P |  |  |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |  |  |
|        | E                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | C                                                                        | A                                                                                                                                                                                                                                                                                                                       |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| 23.    | <p><b>10 sätt; 10</b></p> <p>Påbörjad lösning som t.ex. visar minst fyra korrekta kombinationer.</p> <p>Använder systematisk metod eller visar minst åtta kombinationer.</p> <p>Löser problemet och ger korrekt svar.</p> <p> Till uppgiften finns bedömda elevarbeten.</p>                                                                                                                                                                                                                                                                                                                                | (1/2/0)<br><br>+E <sub>P</sub><br><br>+C <sub>M</sub><br>+C <sub>P</sub> | <table border="1"> <tr><td></td><td>E</td><td>C</td><td>A</td></tr> <tr><td>P</td><td></td><td></td><td></td></tr> <tr><td>B</td><td></td><td></td><td></td></tr> <tr><td>M</td><td></td><td></td><td></td></tr> <tr><td>R</td><td></td><td></td><td></td></tr> <tr><td>K</td><td></td><td></td><td></td></tr> </table> |  | E | C | A | P |  |  |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |  |  |
|        | E                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | C                                                                        | A                                                                                                                                                                                                                                                                                                                       |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                          |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |

|        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                               |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---|---|---|---|--|--|--|---|--|--|--|---|--|--|--|---|--|--|--|---|--|--|--|
| 24.    | <p><b>Nej, det stämmer inte. Motivering där det framgår att lådans volym är t.ex. mer än 9 gånger större eller 27 gånger större.</b></p> <p>Påbörjad lösning, t.ex. anger längdmått för båda lådorna eller resonerar om volymskala.</p> <p>Resonemang och/eller beräkningar som visar att 9 gånger större är felaktigt.</p> <p>Resonemanget är välgrundat och underbyggt.</p> <p> Till uppgiften finns bedömda elevarbeten.</p>                                                                                                                                                                             | <p><b>(0/2/1)</b></p> <p>+C<sub>P</sub></p> <p>+C<sub>R</sub></p> <p>+A<sub>R</sub></p>                       | <table border="1"> <tr><td></td><td>E</td><td>C</td><td>A</td></tr> <tr><td>P</td><td></td><td></td><td></td></tr> <tr><td>B</td><td></td><td></td><td></td></tr> <tr><td>M</td><td></td><td></td><td></td></tr> <tr><td>R</td><td></td><td></td><td></td></tr> <tr><td>K</td><td></td><td></td><td></td></tr> </table> |  | E | C | A | P |  |  |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |  |  |
|        | E                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | C                                                                                                             | A                                                                                                                                                                                                                                                                                                                       |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                               |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                               |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                               |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                               |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                               |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| 25. a) | <p><b>31 %; 31,2 %; 31,22 %</b></p> <p>Lösning som relaterar till rätt helhet.</p> <p>Tydlig redovisning med godtagbart svar.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p><b>(0/2/0)</b></p> <p>+C<sub>B</sub></p> <p>+C<sub>K</sub></p>                                             | <table border="1"> <tr><td></td><td>E</td><td>C</td><td>A</td></tr> <tr><td>P</td><td></td><td></td><td></td></tr> <tr><td>B</td><td></td><td></td><td></td></tr> <tr><td>M</td><td></td><td></td><td></td></tr> <tr><td>R</td><td></td><td></td><td></td></tr> <tr><td>K</td><td></td><td></td><td></td></tr> </table> |  | E | C | A | P |  |  |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |  |  |
|        | E                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | C                                                                                                             | A                                                                                                                                                                                                                                                                                                                       |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                               |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                               |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                               |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                               |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                               |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| b)     | <p><b>6 821 miljoner kronor</b></p> <p>Lösning som visar kunskap om upprepade procentuella ökningar.</p> <p>Tydlig redovisning med godtagbart svar.</p> <p>Använder förändringsfaktor.</p> <p>Lösningen visar dessutom en ändamålsenlig eller effektiv metod vid hantering av förändringsfaktor.</p> <p> Till uppgiften finns bedömda elevarbeten.</p>                                                                                                                                                                                                                                                    | <p><b>(0/2/2)</b></p> <p>+C<sub>B</sub></p> <p>+C<sub>K</sub></p> <p>+A<sub>M</sub></p> <p>+A<sub>M</sub></p> | <table border="1"> <tr><td></td><td>E</td><td>C</td><td>A</td></tr> <tr><td>P</td><td></td><td></td><td></td></tr> <tr><td>B</td><td></td><td></td><td></td></tr> <tr><td>M</td><td></td><td></td><td></td></tr> <tr><td>R</td><td></td><td></td><td></td></tr> <tr><td>K</td><td></td><td></td><td></td></tr> </table> |  | E | C | A | P |  |  |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |  |  |
|        | E                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | C                                                                                                             | A                                                                                                                                                                                                                                                                                                                       |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                               |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                               |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                               |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                               |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                               |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| 26.    | <p><b>Svar 18 % (18,2 %)</b></p> <p>Påbörjad lösning, t.ex. beräknar någon procentuell förändring eller det gamla kilopriset.</p> <p>Beräknar det <i>nya priset</i>, den <i>nya vikten</i> samt det <i>gamla kilopriset</i>.</p> <p>Bestämmer kilopriset för den nya chokladkakan godtagbart.</p> <p>Löser hela problemet med lämpliga avrundningar och godtagbart svar. (Om eleven beräknar förändringen av kilopris genom att beräkna kvoten av förändringsfaktorerna <math>\frac{1,3}{1,1}</math> erhålls även C<sub>B</sub> och C<sub>P</sub>.)</p> <p> Till uppgiften finns bedömda elevarbeten.</p> | <p><b>(1/2/1)</b></p> <p>+E<sub>P</sub></p> <p>+C<sub>B</sub></p> <p>+C<sub>P</sub></p> <p>+A<sub>P</sub></p> | <table border="1"> <tr><td></td><td>E</td><td>C</td><td>A</td></tr> <tr><td>P</td><td></td><td></td><td></td></tr> <tr><td>B</td><td></td><td></td><td></td></tr> <tr><td>M</td><td></td><td></td><td></td></tr> <tr><td>R</td><td></td><td></td><td></td></tr> <tr><td>K</td><td></td><td></td><td></td></tr> </table> |  | E | C | A | P |  |  |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |  |  |
|        | E                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | C                                                                                                             | A                                                                                                                                                                                                                                                                                                                       |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                               |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                               |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                               |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                               |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                               |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |

|        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                       |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---|---|---|---|--|--|--|---|--|--|--|---|--|--|--|---|--|--|--|---|--|--|--|
| 27. a) | <b><math>x</math> är antalet barn; barn</b><br>Gör en tolkning av ekvationens variabel.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>(1/0/0)</b><br>+ $E_P$                                                             | <table border="1"> <tr><td></td><td>E</td><td>C</td><td>A</td></tr> <tr><td>P</td><td></td><td></td><td></td></tr> <tr><td>B</td><td></td><td></td><td></td></tr> <tr><td>M</td><td></td><td></td><td></td></tr> <tr><td>R</td><td></td><td></td><td></td></tr> <tr><td>K</td><td></td><td></td><td></td></tr> </table> |  | E | C | A | P |  |  |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |  |  |
|        | E                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | C                                                                                     | A                                                                                                                                                                                                                                                                                                                       |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                       |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                       |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                       |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                       |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                       |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| b)     | <b><math>x = 38</math></b><br>Visar korrekt multiplikation med parentes eller löser ekvationen enbart med prövning.<br>Korrekt användning av likhetstecknet vid ekvationslösning.<br>Tydlig ekvationslösning med korrekt svar.                                                                                                                                                                                                                                                                                                                                            | <b>(0/2/1)</b><br>+ $C_M$<br><br>+ $C_M$<br><br>+ $A_M$                               | <table border="1"> <tr><td></td><td>E</td><td>C</td><td>A</td></tr> <tr><td>P</td><td></td><td></td><td></td></tr> <tr><td>B</td><td></td><td></td><td></td></tr> <tr><td>M</td><td></td><td></td><td></td></tr> <tr><td>R</td><td></td><td></td><td></td></tr> <tr><td>K</td><td></td><td></td><td></td></tr> </table> |  | E | C | A | P |  |  |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |  |  |
|        | E                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | C                                                                                     | A                                                                                                                                                                                                                                                                                                                       |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                       |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                       |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                       |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                       |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                       |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| c)     | <b>38 barn och 54 vuxna</b><br>Tolkar ekvationen så att antalet vuxna bestäms till 54.<br>Redovisad korrekt beräkning av antalet vuxna.<br>Tydlig redovisning av hela problemet (a–c) med korrekt matematiskt språk.<br> Till uppgiften finns bedömda elevarbeten.                                                                                                                                                                                                                      | <b>(0/2/1)</b><br>+ $C_P$<br><br>+ $C_K$<br><br>+ $A_K$                               | <table border="1"> <tr><td></td><td>E</td><td>C</td><td>A</td></tr> <tr><td>P</td><td></td><td></td><td></td></tr> <tr><td>B</td><td></td><td></td><td></td></tr> <tr><td>M</td><td></td><td></td><td></td></tr> <tr><td>R</td><td></td><td></td><td></td></tr> <tr><td>K</td><td></td><td></td><td></td></tr> </table> |  | E | C | A | P |  |  |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |  |  |
|        | E                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | C                                                                                     | A                                                                                                                                                                                                                                                                                                                       |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                       |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                       |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                       |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                       |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                       |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| 28.    | <b>Avgift 80 kr per gång och medlemskort 100 kr</b><br>Påbörjad lösning som visar förståelse för betalningssystemet.<br>Visar en metod som bygger på sambandet mellan differensen i betalning och differensen i antalet besök eller tecknar Amys och Liams kostnader algebraiskt.<br>Lösar hela problemet med korrekt svar.<br>Tydlig och välstrukturerad lösning med korrekt matematiskt språk.<br>Använder dessutom en generell metod.<br> Till uppgiften finns bedömda elevarbeten. | <b>(0/2/3)</b><br>+ $C_P$<br><br>+ $C_M$<br><br>+ $A_P$<br><br>+ $A_K$<br><br>+ $A_M$ | <table border="1"> <tr><td></td><td>E</td><td>C</td><td>A</td></tr> <tr><td>P</td><td></td><td></td><td></td></tr> <tr><td>B</td><td></td><td></td><td></td></tr> <tr><td>M</td><td></td><td></td><td></td></tr> <tr><td>R</td><td></td><td></td><td></td></tr> <tr><td>K</td><td></td><td></td><td></td></tr> </table> |  | E | C | A | P |  |  |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |  |  |
|        | E                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | C                                                                                     | A                                                                                                                                                                                                                                                                                                                       |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                       |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                       |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                       |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                       |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                       |                                                                                                                                                                                                                                                                                                                         |  |   |   |   |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |  |  |

## Bedömda elevarbeten till Delprov D

Bedömda elevarbeten till uppgift 18

Max (3/0/0)

|                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |  |  |   |  |  |  |   |   |  |  |   |  |  |  |   |   |  |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|--|--|---|--|--|--|---|---|--|--|---|--|--|--|---|---|--|--|
| <div>Elevarbete 1</div> <div><math>12 + 8 + 5 = 25</math><br/><math>\frac{400}{25} = 16</math></div>                                                                                                                                                                                                                                       | <div>1/0/0</div> <div><table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td>X</td><td></td><td></td></tr><tr><td>B</td><td></td><td></td><td></td></tr><tr><td>M</td><td></td><td></td><td></td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>K</td><td></td><td></td><td></td></tr></table></div>   |   | E | C | A | P | X |  |  | B |  |  |  | M |   |  |  | R |  |  |  | K |   |  |  |
|                                                                                                                                                                                                                                                                                                                                            | E                                                                                                                                                                                                                                                                                                                                    | C | A |   |   |   |   |  |  |   |  |  |  |   |   |  |  |   |  |  |  |   |   |  |  |
| P                                                                                                                                                                                                                                                                                                                                          | X                                                                                                                                                                                                                                                                                                                                    |   |   |   |   |   |   |  |  |   |  |  |  |   |   |  |  |   |  |  |  |   |   |  |  |
| B                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |  |  |   |  |  |  |   |   |  |  |   |  |  |  |   |   |  |  |
| M                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |  |  |   |  |  |  |   |   |  |  |   |  |  |  |   |   |  |  |
| R                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |  |  |   |  |  |  |   |   |  |  |   |  |  |  |   |   |  |  |
| K                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |  |  |   |  |  |  |   |   |  |  |   |  |  |  |   |   |  |  |
| <div>Elevarbete 2</div> <div><math>5 \cdot 16 = 80</math><br/><math>8 \cdot 16 = 128</math><br/><math>12 \cdot 16 = 192</math><br/><div><math>\begin{array}{r} 21 \\ 80 \\ 128 \\ + 192 \\ \hline 400 \end{array}</math></div><br/>Svar: 80 apelsiner<br/>128 bananer<br/>192 äpplen</div>                                                 | <div>2/0/0</div> <div><table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td>X</td><td></td><td></td></tr><tr><td>B</td><td></td><td></td><td></td></tr><tr><td>M</td><td>X</td><td></td><td></td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>K</td><td></td><td></td><td></td></tr></table></div>  |   | E | C | A | P | X |  |  | B |  |  |  | M | X |  |  | R |  |  |  | K |   |  |  |
|                                                                                                                                                                                                                                                                                                                                            | E                                                                                                                                                                                                                                                                                                                                    | C | A |   |   |   |   |  |  |   |  |  |  |   |   |  |  |   |  |  |  |   |   |  |  |
| P                                                                                                                                                                                                                                                                                                                                          | X                                                                                                                                                                                                                                                                                                                                    |   |   |   |   |   |   |  |  |   |  |  |  |   |   |  |  |   |  |  |  |   |   |  |  |
| B                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |  |  |   |  |  |  |   |   |  |  |   |  |  |  |   |   |  |  |
| M                                                                                                                                                                                                                                                                                                                                          | X                                                                                                                                                                                                                                                                                                                                    |   |   |   |   |   |   |  |  |   |  |  |  |   |   |  |  |   |  |  |  |   |   |  |  |
| R                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |  |  |   |  |  |  |   |   |  |  |   |  |  |  |   |   |  |  |
| K                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |  |  |   |  |  |  |   |   |  |  |   |  |  |  |   |   |  |  |
| <div>Elevarbete 3</div> <div><math>5 + 8 + 12 = 25</math><br/><math>400/25 = 16</math><br/><math>\left. \begin{array}{l} 5 \cdot 16 = 80 \\ 8 \cdot 16 = 128 \\ 12 \cdot 16 = 192 \end{array} \right\} = 400 \text{ frukter}</math><br/>Svar: 80 apelsiner, 128 bananer, 192 äpplen</div>                                                  | <div>3/0/0</div> <div><table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td>X</td><td></td><td></td></tr><tr><td>B</td><td></td><td></td><td></td></tr><tr><td>M</td><td>X</td><td></td><td></td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>K</td><td>X</td><td></td><td></td></tr></table></div> |   | E | C | A | P | X |  |  | B |  |  |  | M | X |  |  | R |  |  |  | K | X |  |  |
|                                                                                                                                                                                                                                                                                                                                            | E                                                                                                                                                                                                                                                                                                                                    | C | A |   |   |   |   |  |  |   |  |  |  |   |   |  |  |   |  |  |  |   |   |  |  |
| P                                                                                                                                                                                                                                                                                                                                          | X                                                                                                                                                                                                                                                                                                                                    |   |   |   |   |   |   |  |  |   |  |  |  |   |   |  |  |   |  |  |  |   |   |  |  |
| B                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |  |  |   |  |  |  |   |   |  |  |   |  |  |  |   |   |  |  |
| M                                                                                                                                                                                                                                                                                                                                          | X                                                                                                                                                                                                                                                                                                                                    |   |   |   |   |   |   |  |  |   |  |  |  |   |   |  |  |   |  |  |  |   |   |  |  |
| R                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |  |  |   |  |  |  |   |   |  |  |   |  |  |  |   |   |  |  |
| K                                                                                                                                                                                                                                                                                                                                          | X                                                                                                                                                                                                                                                                                                                                    |   |   |   |   |   |   |  |  |   |  |  |  |   |   |  |  |   |  |  |  |   |   |  |  |
| <div>Elevarbete 4</div> <div><math>5 + 8 + 12 = 25</math><br/><math>\frac{5}{25} = 0,2</math>    <math>\frac{8}{25} = 0,32</math>    <math>\frac{12}{25} = 0,48</math><br/><math>400 \cdot 0,2 = 80</math>    <math>400 \cdot 0,32 = 128</math>    <math>400 \cdot 0,48 = 192</math><br/>Svar: 80 apelsiner, 128 bananer, 192 äpplen</div> | <div>3/0/0</div> <div><table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td>X</td><td></td><td></td></tr><tr><td>B</td><td></td><td></td><td></td></tr><tr><td>M</td><td>X</td><td></td><td></td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>K</td><td>X</td><td></td><td></td></tr></table></div> |   | E | C | A | P | X |  |  | B |  |  |  | M | X |  |  | R |  |  |  | K | X |  |  |
|                                                                                                                                                                                                                                                                                                                                            | E                                                                                                                                                                                                                                                                                                                                    | C | A |   |   |   |   |  |  |   |  |  |  |   |   |  |  |   |  |  |  |   |   |  |  |
| P                                                                                                                                                                                                                                                                                                                                          | X                                                                                                                                                                                                                                                                                                                                    |   |   |   |   |   |   |  |  |   |  |  |  |   |   |  |  |   |  |  |  |   |   |  |  |
| B                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |  |  |   |  |  |  |   |   |  |  |   |  |  |  |   |   |  |  |
| M                                                                                                                                                                                                                                                                                                                                          | X                                                                                                                                                                                                                                                                                                                                    |   |   |   |   |   |   |  |  |   |  |  |  |   |   |  |  |   |  |  |  |   |   |  |  |
| R                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |  |  |   |  |  |  |   |   |  |  |   |  |  |  |   |   |  |  |
| K                                                                                                                                                                                                                                                                                                                                          | X                                                                                                                                                                                                                                                                                                                                    |   |   |   |   |   |   |  |  |   |  |  |  |   |   |  |  |   |  |  |  |   |   |  |  |

|                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |   |   |  |   |  |  |  |   |   |  |  |   |  |  |  |   |  |  |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|--|---|--|--|--|---|---|--|--|---|--|--|--|---|--|--|--|
| Elevarbete 1                                                                                                                                                                                                                                                                                                                  | 1/0/0<br><table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td>X</td><td></td><td></td></tr><tr><td>B</td><td></td><td></td><td></td></tr><tr><td>M</td><td></td><td></td><td></td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>K</td><td></td><td></td><td></td></tr></table>   |   | E | C | A | P | X |   |  | B |  |  |  | M |   |  |  | R |  |  |  | K |  |  |  |
|                                                                                                                                                                                                                                                                                                                               | E                                                                                                                                                                                                                                                                                                                 | C | A |   |   |   |   |   |  |   |  |  |  |   |   |  |  |   |  |  |  |   |  |  |  |
| P                                                                                                                                                                                                                                                                                                                             | X                                                                                                                                                                                                                                                                                                                 |   |   |   |   |   |   |   |  |   |  |  |  |   |   |  |  |   |  |  |  |   |  |  |  |
| B                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |   |   |  |   |  |  |  |   |   |  |  |   |  |  |  |   |  |  |  |
| M                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |   |   |  |   |  |  |  |   |   |  |  |   |  |  |  |   |  |  |  |
| R                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |   |   |  |   |  |  |  |   |   |  |  |   |  |  |  |   |  |  |  |
| K                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |   |   |  |   |  |  |  |   |   |  |  |   |  |  |  |   |  |  |  |
| <p>10 st bröd kan han baka. Då kostar det <math>17 \cdot 10 = 170</math> kr.</p> <p>Om han köpte 10 bröd i affären så kostar det <math>29 \cdot 10 = 290</math> kr.</p> <p><math>170 \xrightarrow{30} 200 \xrightarrow{90} 290</math></p> <p><math>30 + 90 = 120</math></p> <p>Svar: Minst 10 bröd, då tjänar han 120 kr.</p> |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |   |   |  |   |  |  |  |   |   |  |  |   |  |  |  |   |  |  |  |
| Elevarbete 2                                                                                                                                                                                                                                                                                                                  | 2/0/0<br><table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td>X</td><td></td><td></td></tr><tr><td>B</td><td></td><td></td><td></td></tr><tr><td>M</td><td>X</td><td></td><td></td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>K</td><td></td><td></td><td></td></tr></table>  |   | E | C | A | P | X |   |  | B |  |  |  | M | X |  |  | R |  |  |  | K |  |  |  |
|                                                                                                                                                                                                                                                                                                                               | E                                                                                                                                                                                                                                                                                                                 | C | A |   |   |   |   |   |  |   |  |  |  |   |   |  |  |   |  |  |  |   |  |  |  |
| P                                                                                                                                                                                                                                                                                                                             | X                                                                                                                                                                                                                                                                                                                 |   |   |   |   |   |   |   |  |   |  |  |  |   |   |  |  |   |  |  |  |   |  |  |  |
| B                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |   |   |  |   |  |  |  |   |   |  |  |   |  |  |  |   |  |  |  |
| M                                                                                                                                                                                                                                                                                                                             | X                                                                                                                                                                                                                                                                                                                 |   |   |   |   |   |   |   |  |   |  |  |  |   |   |  |  |   |  |  |  |   |  |  |  |
| R                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |   |   |  |   |  |  |  |   |   |  |  |   |  |  |  |   |  |  |  |
| K                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |   |   |  |   |  |  |  |   |   |  |  |   |  |  |  |   |  |  |  |
| <p><math>29 - 17 = 12</math>      <math>\frac{495}{12} = 41,25</math>.</p> <p>Det tar 41 bröd innan man får tillbaka pengarna.</p>                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |   |   |  |   |  |  |  |   |   |  |  |   |  |  |  |   |  |  |  |
| Elevarbete 3                                                                                                                                                                                                                                                                                                                  | 2/1/0<br><table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td>X</td><td>X</td><td></td></tr><tr><td>B</td><td></td><td></td><td></td></tr><tr><td>M</td><td>X</td><td></td><td></td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>K</td><td></td><td></td><td></td></tr></table> |   | E | C | A | P | X | X |  | B |  |  |  | M | X |  |  | R |  |  |  | K |  |  |  |
|                                                                                                                                                                                                                                                                                                                               | E                                                                                                                                                                                                                                                                                                                 | C | A |   |   |   |   |   |  |   |  |  |  |   |   |  |  |   |  |  |  |   |  |  |  |
| P                                                                                                                                                                                                                                                                                                                             | X                                                                                                                                                                                                                                                                                                                 | X |   |   |   |   |   |   |  |   |  |  |  |   |   |  |  |   |  |  |  |   |  |  |  |
| B                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |   |   |  |   |  |  |  |   |   |  |  |   |  |  |  |   |  |  |  |
| M                                                                                                                                                                                                                                                                                                                             | X                                                                                                                                                                                                                                                                                                                 |   |   |   |   |   |   |   |  |   |  |  |  |   |   |  |  |   |  |  |  |   |  |  |  |
| R                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |   |   |  |   |  |  |  |   |   |  |  |   |  |  |  |   |  |  |  |
| K                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |   |   |  |   |  |  |  |   |   |  |  |   |  |  |  |   |  |  |  |
| <p><math>29 - 17 = 12</math></p> <p><math>495 / 12 = 41,25</math></p> <p>Svar: 42 bröd</p>                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |   |   |  |   |  |  |  |   |   |  |  |   |  |  |  |   |  |  |  |

|                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                       |   |   |   |   |   |   |   |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |   |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|--|---|--|--|--|---|--|--|--|---|--|--|--|---|--|---|--|
| <p>Elevarbete 1</p> <p>□ □ ○ ○ - 74 kr</p> <p>□ ○ ○ - 53 kr</p> <p>74 - 53 = 21 kr 1 bulle - 21 kr 21 · 2 = 42 74 - 42 = 32</p>                                                                                                                                                                                                   | <p>1/0/0</p> <table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td>X</td><td></td><td></td></tr><tr><td>B</td><td></td><td></td><td></td></tr><tr><td>M</td><td></td><td></td><td></td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>K</td><td></td><td></td><td></td></tr></table>   |   | E | C | A | P | X |   |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |   |  |
|                                                                                                                                                                                                                                                                                                                                   | E                                                                                                                                                                                                                                                                                                                     | C | A |   |   |   |   |   |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |   |  |
| P                                                                                                                                                                                                                                                                                                                                 | X                                                                                                                                                                                                                                                                                                                     |   |   |   |   |   |   |   |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |   |  |
| B                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                       |   |   |   |   |   |   |   |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |   |  |
| M                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                       |   |   |   |   |   |   |   |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |   |  |
| R                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                       |   |   |   |   |   |   |   |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |   |  |
| K                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                       |   |   |   |   |   |   |   |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |   |  |
| <p>Elevarbete 2</p> <p>Amy = 2b + 2j = 74</p> <p>Kund = 2b + j = 53</p>                                                                                                                                                                                                                                                           | <p>1/0/0</p> <table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td>X</td><td></td><td></td></tr><tr><td>B</td><td></td><td></td><td></td></tr><tr><td>M</td><td></td><td></td><td></td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>K</td><td></td><td></td><td></td></tr></table>   |   | E | C | A | P | X |   |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |   |  |
|                                                                                                                                                                                                                                                                                                                                   | E                                                                                                                                                                                                                                                                                                                     | C | A |   |   |   |   |   |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |   |  |
| P                                                                                                                                                                                                                                                                                                                                 | X                                                                                                                                                                                                                                                                                                                     |   |   |   |   |   |   |   |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |   |  |
| B                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                       |   |   |   |   |   |   |   |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |   |  |
| M                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                       |   |   |   |   |   |   |   |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |   |  |
| R                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                       |   |   |   |   |   |   |   |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |   |  |
| K                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                       |   |   |   |   |   |   |   |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |   |  |
| <p>Elevarbete 3</p> <p>16 + 16 + 21 + 21 = 74</p> <p>16 + 16 + 21 = 53</p> <p>Bulle kostar <u>16</u> kr</p>                                                                                                                                                                                                                       | <p>1/1/0</p> <table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td>X</td><td>X</td><td></td></tr><tr><td>B</td><td></td><td></td><td></td></tr><tr><td>M</td><td></td><td></td><td></td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>K</td><td></td><td></td><td></td></tr></table>  |   | E | C | A | P | X | X |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  |   |  |
|                                                                                                                                                                                                                                                                                                                                   | E                                                                                                                                                                                                                                                                                                                     | C | A |   |   |   |   |   |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |   |  |
| P                                                                                                                                                                                                                                                                                                                                 | X                                                                                                                                                                                                                                                                                                                     | X |   |   |   |   |   |   |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |   |  |
| B                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                       |   |   |   |   |   |   |   |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |   |  |
| M                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                       |   |   |   |   |   |   |   |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |   |  |
| R                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                       |   |   |   |   |   |   |   |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |   |  |
| K                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                       |   |   |   |   |   |   |   |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |   |  |
| <p>Elevarbete 4</p> <p>x är bulle y är juice</p> <p>2x + 2y = 74</p> <p>1x + 2y = 53</p> <p>74 - 53 = 21</p> <p>x = 21</p> <p>2(21) + 2y = 74</p> <p>2y = 32</p> <p>y = 16 Svar: En bulle kostar 16 kr.</p>                                                                                                                       | <p>1/2/0</p> <table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td>X</td><td>X</td><td></td></tr><tr><td>B</td><td></td><td></td><td></td></tr><tr><td>M</td><td></td><td></td><td></td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>K</td><td></td><td>X</td><td></td></tr></table> |   | E | C | A | P | X | X |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  | X |  |
|                                                                                                                                                                                                                                                                                                                                   | E                                                                                                                                                                                                                                                                                                                     | C | A |   |   |   |   |   |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |   |  |
| P                                                                                                                                                                                                                                                                                                                                 | X                                                                                                                                                                                                                                                                                                                     | X |   |   |   |   |   |   |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |   |  |
| B                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                       |   |   |   |   |   |   |   |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |   |  |
| M                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                       |   |   |   |   |   |   |   |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |   |  |
| R                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                       |   |   |   |   |   |   |   |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |   |  |
| K                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                       | X |   |   |   |   |   |   |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |   |  |
| <p>Elevarbete 5</p> <p>Amy betalar för 2 bullar &amp; 2 glas juice = 74 kr</p> <p>Kunden före betalar 53 kr för 2 bullar &amp; 1 glas juice.</p> <p>1 glas juice = 21 kr (74 - 53 = 21)</p> <p>2 glas juice = 21 · 2 = 42 kr</p> <p>74 - 42 = 32 kr = 2 bullar</p> <p>Svar: En bulle kostar 16 kr (<math>\frac{32}{2}</math>)</p> | <p>1/2/0</p> <table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td>X</td><td>X</td><td></td></tr><tr><td>B</td><td></td><td></td><td></td></tr><tr><td>M</td><td></td><td></td><td></td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>K</td><td></td><td>X</td><td></td></tr></table> |   | E | C | A | P | X | X |  | B |  |  |  | M |  |  |  | R |  |  |  | K |  | X |  |
|                                                                                                                                                                                                                                                                                                                                   | E                                                                                                                                                                                                                                                                                                                     | C | A |   |   |   |   |   |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |   |  |
| P                                                                                                                                                                                                                                                                                                                                 | X                                                                                                                                                                                                                                                                                                                     | X |   |   |   |   |   |   |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |   |  |
| B                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                       |   |   |   |   |   |   |   |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |   |  |
| M                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                       |   |   |   |   |   |   |   |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |   |  |
| R                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                       |   |   |   |   |   |   |   |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |   |  |
| K                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                       | X |   |   |   |   |   |   |  |   |  |  |  |   |  |  |  |   |  |  |  |   |  |   |  |

|              |                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |  |   |  |   |   |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|--|---|--|---|---|--|--|---|--|---|--|---|--|--|--|---|--|--|--|
| Elevarbete 1 | <div><math display="block">\frac{1}{8} + \frac{1}{12} = \frac{7}{12} = 15 \quad \frac{15}{7} = 2,14 \quad 2,14 \cdot 12 = 25,7 \approx</math><math display="block">\approx \underline{26} \text{ salamikonver}</math></div>                                                                               | <div>1/0/0</div> <table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td></td><td></td><td></td></tr><tr><td>B</td><td>X</td><td></td><td></td></tr><tr><td>M</td><td></td><td></td><td></td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>K</td><td></td><td></td><td></td></tr></table>   |   | E | C | A | P |  |   |  | B | X |  |  | M |  |   |  | R |  |  |  | K |  |  |  |
|              | E                                                                                                                                                                                                                                                                                                         | C                                                                                                                                                                                                                                                                                                                         | A |   |   |   |   |  |   |  |   |   |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| P            |                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |  |   |  |   |   |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| B            | X                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |  |   |  |   |   |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| M            |                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |  |   |  |   |   |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| R            |                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |  |   |  |   |   |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| K            |                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |  |   |  |   |   |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| Elevarbete 2 | <div><math display="block">\frac{5}{12} = 15 \text{ konver}</math><p>Erik: <math>\frac{1}{3} = \frac{4}{12}</math><br/>Liam: <math>\frac{1}{4} = \frac{3}{12}</math>      Svar: ?</p></div>                              | <div>1/1/0</div> <table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td></td><td></td><td></td></tr><tr><td>B</td><td>X</td><td></td><td></td></tr><tr><td>M</td><td></td><td>X</td><td></td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>K</td><td></td><td></td><td></td></tr></table>  |   | E | C | A | P |  |   |  | B | X |  |  | M |  | X |  | R |  |  |  | K |  |  |  |
|              | E                                                                                                                                                                                                                                                                                                         | C                                                                                                                                                                                                                                                                                                                         | A |   |   |   |   |  |   |  |   |   |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| P            |                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |  |   |  |   |   |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| B            | X                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |  |   |  |   |   |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| M            |                                                                                                                                                                                                                                                                                                           | X                                                                                                                                                                                                                                                                                                                         |   |   |   |   |   |  |   |  |   |   |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| R            |                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |  |   |  |   |   |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| K            |                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |  |   |  |   |   |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| Elevarbete 3 | <div><p>Erik sålde <math>\frac{1}{3} \approx 33,3\%</math><br/>Liam sålde <math>\frac{1}{4} = 25\%</math><br/><math>100 - 58,3 = 41,7\% \approx 42\%</math><br/>15 konver = 42%      <math>\frac{15}{42} = 0,35 \text{ konver} = 1\%</math><br/><math>0,35 \cdot 100 = 35 \text{ konver}</math></p></div> | <div>1/1/0</div> <table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td></td><td></td><td></td></tr><tr><td>B</td><td>X</td><td></td><td></td></tr><tr><td>M</td><td></td><td>X</td><td></td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>K</td><td></td><td></td><td></td></tr></table>  |   | E | C | A | P |  |   |  | B | X |  |  | M |  | X |  | R |  |  |  | K |  |  |  |
|              | E                                                                                                                                                                                                                                                                                                         | C                                                                                                                                                                                                                                                                                                                         | A |   |   |   |   |  |   |  |   |   |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| P            |                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |  |   |  |   |   |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| B            | X                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |  |   |  |   |   |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| M            |                                                                                                                                                                                                                                                                                                           | X                                                                                                                                                                                                                                                                                                                         |   |   |   |   |   |  |   |  |   |   |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| R            |                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |  |   |  |   |   |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| K            |                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |  |   |  |   |   |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| Elevarbete 4 | <div><math display="block">\frac{1}{3} + \frac{1}{4} = \frac{4}{12} + \frac{3}{12} = \frac{7}{12}</math><p><math>\frac{5}{12}</math> av lådan = 15<br/><math>\frac{1}{12}</math> av lådan = 3<br/><math>\frac{12}{12}</math> av lådan = <math>3 \cdot 12 = 36</math> konver</p></div>                     | <div>1/2/0</div> <table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td></td><td>X</td><td></td></tr><tr><td>B</td><td>X</td><td></td><td></td></tr><tr><td>M</td><td></td><td>X</td><td></td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>K</td><td></td><td></td><td></td></tr></table> |   | E | C | A | P |  | X |  | B | X |  |  | M |  | X |  | R |  |  |  | K |  |  |  |
|              | E                                                                                                                                                                                                                                                                                                         | C                                                                                                                                                                                                                                                                                                                         | A |   |   |   |   |  |   |  |   |   |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| P            |                                                                                                                                                                                                                                                                                                           | X                                                                                                                                                                                                                                                                                                                         |   |   |   |   |   |  |   |  |   |   |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| B            | X                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |  |   |  |   |   |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| M            |                                                                                                                                                                                                                                                                                                           | X                                                                                                                                                                                                                                                                                                                         |   |   |   |   |   |  |   |  |   |   |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| R            |                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |  |   |  |   |   |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| K            |                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |  |   |  |   |   |  |  |   |  |   |  |   |  |  |  |   |  |  |  |

## Elevarbete 1

Å = Ärtor      Å G P    PTM    PGM  
 G = Gurka      GPT    MÅP    MTÅ  
 P = Paprika  
 T = Tomat  
 M = Majs

Svar: 6 st. olika

1/0/0

|   |   |   |   |
|---|---|---|---|
|   | E | C | A |
| P | X |   |   |
| B |   |   |   |
| M |   |   |   |
| R |   |   |   |
| K |   |   |   |

## Elevarbete 2

| Ärtor | Gurka | Paprika | Tomat | Majs |
|-------|-------|---------|-------|------|
| X     | X     | X       |       |      |
| X     | X     |         | X     |      |
| X     | X     |         |       | X    |
| X     |       | X       | X     |      |
| X     |       | X       |       | X    |
| X     |       |         | X     | X    |
|       |       |         |       |      |
|       |       |         |       |      |

Man kan göra 6 olika sallader

Kommentar: Använder en systematisk metod vid val av grönsaker.

1/1/0

|   |   |   |   |
|---|---|---|---|
|   | E | C | A |
| P | X |   |   |
| B |   |   |   |
| M |   | X |   |
| R |   |   |   |
| K |   |   |   |

## Elevarbete 3

Å = ärtor    G = gurka    P = paprika    T = tomat    M = majs  
 ÅGP    GPT    PTM  
 ÅGT    GPM  
 ÅGM    GTM  
 ÅPT  
 ÅPM  
 ÅTM

= 10 olika sätt

1/2/0

|   |   |   |   |
|---|---|---|---|
|   | E | C | A |
| P | X | X |   |
| B |   |   |   |
| M |   | X |   |
| R |   |   |   |
| K |   |   |   |

## Elevarbete 4

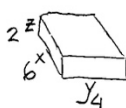
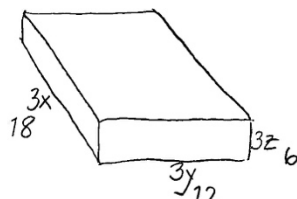
Å    G    P    T    M  
 \_\_\_\_\_  
 \_\_\_\_\_  
 \_\_\_\_\_  
 \_\_\_\_\_  
 \_\_\_\_\_  
 \_\_\_\_\_  
 \_\_\_\_\_  
 \_\_\_\_\_

10 olika sallader

1/2/0

|   |   |   |   |
|---|---|---|---|
|   | E | C | A |
| P | X | X |   |
| B |   |   |   |
| M |   | X |   |
| R |   |   |   |
| K |   |   |   |



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |  |   |  |   |  |  |  |   |  |  |  |   |  |   |   |   |  |  |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|--|---|--|---|--|--|--|---|--|--|--|---|--|---|---|---|--|--|--|
| <div>Elevarbete 1</div> <div>Lilla exemplet: <math>2 \cdot 2 \cdot 4 = 16 \text{ cm}^3</math><br/>Stora exemplet: <math>6 \cdot 6 \cdot 12 = 432 \text{ cm}^3</math></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <div>0/1/0</div> <table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td></td><td>X</td><td></td></tr><tr><td>B</td><td></td><td></td><td></td></tr><tr><td>M</td><td></td><td></td><td></td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>K</td><td></td><td></td><td></td></tr></table>   |   | E | C | A | P |  | X |  | B |  |  |  | M |  |  |  | R |  |   |   | K |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | E                                                                                                                                                                                                                                                                                                                         | C | A |   |   |   |  |   |  |   |  |  |  |   |  |  |  |   |  |   |   |   |  |  |  |
| P                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                           | X |   |   |   |   |  |   |  |   |  |  |  |   |  |  |  |   |  |   |   |   |  |  |  |
| B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |  |   |  |   |  |  |  |   |  |  |  |   |  |   |   |   |  |  |  |
| M                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |  |   |  |   |  |  |  |   |  |  |  |   |  |   |   |   |  |  |  |
| R                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |  |   |  |   |  |  |  |   |  |  |  |   |  |   |   |   |  |  |  |
| K                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |  |   |  |   |  |  |  |   |  |  |  |   |  |   |   |   |  |  |  |
| <div>Elevarbete 2</div> <div>Nej, <math>3 \cdot 3 \cdot 3 = 27</math> Svar: Den är 27 ggr större.</div> <div>Kommentar: Redovisningen är alltför knapphändig och därmed inte tillräckligt underbyggd.</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <div>0/2/0</div> <table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td></td><td>X</td><td></td></tr><tr><td>B</td><td></td><td></td><td></td></tr><tr><td>M</td><td></td><td></td><td></td></tr><tr><td>R</td><td></td><td>X</td><td></td></tr><tr><td>K</td><td></td><td></td><td></td></tr></table>  |   | E | C | A | P |  | X |  | B |  |  |  | M |  |  |  | R |  | X |   | K |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | E                                                                                                                                                                                                                                                                                                                         | C | A |   |   |   |  |   |  |   |  |  |  |   |  |  |  |   |  |   |   |   |  |  |  |
| P                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                           | X |   |   |   |   |  |   |  |   |  |  |  |   |  |  |  |   |  |   |   |   |  |  |  |
| B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |  |   |  |   |  |  |  |   |  |  |  |   |  |   |   |   |  |  |  |
| M                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |  |   |  |   |  |  |  |   |  |  |  |   |  |   |   |   |  |  |  |
| R                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                           | X |   |   |   |   |  |   |  |   |  |  |  |   |  |  |  |   |  |   |   |   |  |  |  |
| K                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |  |   |  |   |  |  |  |   |  |  |  |   |  |   |   |   |  |  |  |
| <div>Elevarbete 3</div> <div>Om den lilla lådens mått skulle vara =<br/>längd: 2 cm, bredd: 1 cm och höjd: 1 cm skulle<br/>volymen vara: <math>2 \cdot 1 = 2 \text{ cm}^2</math> <math>2 \cdot 1 = 2 \text{ cm}^3</math><br/>Då skulle den stora lådens mått vara =<br/>längd: 6 cm, bredd: 3 cm och höjd 3 cm.<br/>Arean: <math>6 \cdot 3 = 18 \text{ cm}^2</math> Volym: <math>18 \cdot 3 = 54 \text{ cm}^3</math><br/>Svar: Liams påstående är fel eftersom om den lilla<br/>lådens volym skulle vara <math>2 \text{ cm}^3</math> skulle den<br/>stora lådens volym vara <math>18 \text{ cm}^3</math> om den var<br/>9 gånger större eftersom <math>2 \cdot 9 = 18</math>.</div> | <div>0/2/1</div> <table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td></td><td>X</td><td></td></tr><tr><td>B</td><td></td><td></td><td></td></tr><tr><td>M</td><td></td><td></td><td></td></tr><tr><td>R</td><td></td><td>X</td><td>X</td></tr><tr><td>K</td><td></td><td></td><td></td></tr></table> |   | E | C | A | P |  | X |  | B |  |  |  | M |  |  |  | R |  | X | X | K |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | E                                                                                                                                                                                                                                                                                                                         | C | A |   |   |   |  |   |  |   |  |  |  |   |  |  |  |   |  |   |   |   |  |  |  |
| P                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                           | X |   |   |   |   |  |   |  |   |  |  |  |   |  |  |  |   |  |   |   |   |  |  |  |
| B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |  |   |  |   |  |  |  |   |  |  |  |   |  |   |   |   |  |  |  |
| M                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |  |   |  |   |  |  |  |   |  |  |  |   |  |   |   |   |  |  |  |
| R                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                           | X | X |   |   |   |  |   |  |   |  |  |  |   |  |  |  |   |  |   |   |   |  |  |  |
| K                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |  |   |  |   |  |  |  |   |  |  |  |   |  |   |   |   |  |  |  |
| <div>Elevarbete 4</div> <div><div><div><p><math>V = 48</math></p></div><div><p><math>V = 1296</math></p></div></div><div><math>\frac{1296}{48} = 27</math></div><div>Om varje sida blir 3ggr så stor så blir lådens volym 27ggr större eftersom varje sida (3 st som man räknar när man räknar volym) blir 3ggr större. Då blir lådens volym <math>3^3 = 27</math> ggr så stor.</div></div>                                                                                                                   | <div>0/2/1</div> <table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td></td><td>X</td><td></td></tr><tr><td>B</td><td></td><td></td><td></td></tr><tr><td>M</td><td></td><td></td><td></td></tr><tr><td>R</td><td></td><td>X</td><td>X</td></tr><tr><td>K</td><td></td><td></td><td></td></tr></table> |   | E | C | A | P |  | X |  | B |  |  |  | M |  |  |  | R |  | X | X | K |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | E                                                                                                                                                                                                                                                                                                                         | C | A |   |   |   |  |   |  |   |  |  |  |   |  |  |  |   |  |   |   |   |  |  |  |
| P                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                           | X |   |   |   |   |  |   |  |   |  |  |  |   |  |  |  |   |  |   |   |   |  |  |  |
| B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |  |   |  |   |  |  |  |   |  |  |  |   |  |   |   |   |  |  |  |
| M                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |  |   |  |   |  |  |  |   |  |  |  |   |  |   |   |   |  |  |  |
| R                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                           | X | X |   |   |   |  |   |  |   |  |  |  |   |  |  |  |   |  |   |   |   |  |  |  |
| K                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |  |   |  |   |  |  |  |   |  |  |  |   |  |   |   |   |  |  |  |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |  |  |  |   |  |   |  |   |  |  |   |   |  |  |  |   |  |   |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|--|--|--|---|--|---|--|---|--|--|---|---|--|--|--|---|--|---|--|
| Elevarbete 1                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0/1/0<br><table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td></td><td></td><td></td></tr><tr><td>B</td><td></td><td>X</td><td></td></tr><tr><td>M</td><td></td><td></td><td>/</td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>K</td><td></td><td></td><td></td></tr></table>  |   | E | C | A | P |  |  |  | B |  | X |  | M |  |  | / | R |  |  |  | K |  |   |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | E                                                                                                                                                                                                                                                                                                                 | C | A |   |   |   |  |  |  |   |  |   |  |   |  |  |   |   |  |  |  |   |  |   |  |
| P                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |  |  |  |   |  |   |  |   |  |  |   |   |  |  |  |   |  |   |  |
| B                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                   | X |   |   |   |   |  |  |  |   |  |   |  |   |  |  |   |   |  |  |  |   |  |   |  |
| M                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                   |   | / |   |   |   |  |  |  |   |  |   |  |   |  |  |   |   |  |  |  |   |  |   |  |
| R                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |  |  |  |   |  |   |  |   |  |  |   |   |  |  |  |   |  |   |  |
| K                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |  |  |  |   |  |   |  |   |  |  |   |   |  |  |  |   |  |   |  |
| <p>7% per år</p> $0,07 \cdot 4863 = 340,41$ $340,41 + 4863 = 5203,41$ $0,07 \cdot 5203,41 = 364,2387$ $364,2387 + 5203,41 \quad \text{orka}$                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |  |  |  |   |  |   |  |   |  |  |   |   |  |  |  |   |  |   |  |
| Elevarbete 2                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0/2/0<br><table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td></td><td></td><td></td></tr><tr><td>B</td><td></td><td>X</td><td></td></tr><tr><td>M</td><td></td><td></td><td>/</td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>K</td><td></td><td>X</td><td></td></tr></table> |   | E | C | A | P |  |  |  | B |  | X |  | M |  |  | / | R |  |  |  | K |  | X |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | E                                                                                                                                                                                                                                                                                                                 | C | A |   |   |   |  |  |  |   |  |   |  |   |  |  |   |   |  |  |  |   |  |   |  |
| P                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |  |  |  |   |  |   |  |   |  |  |   |   |  |  |  |   |  |   |  |
| B                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                   | X |   |   |   |   |  |  |  |   |  |   |  |   |  |  |   |   |  |  |  |   |  |   |  |
| M                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                   |   | / |   |   |   |  |  |  |   |  |   |  |   |  |  |   |   |  |  |  |   |  |   |  |
| R                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |  |  |  |   |  |   |  |   |  |  |   |   |  |  |  |   |  |   |  |
| K                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                   | X |   |   |   |   |  |  |  |   |  |   |  |   |  |  |   |   |  |  |  |   |  |   |  |
| <p>2011: 4863 milj. kr</p> $2012: 4863 \cdot 0,07 = 340,41$ $4863 + 340,41 = 5203,41 \approx 5203 \text{ milj.}$ $2013: 5203 \cdot 0,07 = 364,21$ $5203 + 364,21 = 5567,21 \approx 5567 \text{ milj.}$ $2014: 5567 \cdot 0,07 = 389,69$ $5567 + 389,69 = 5956,69 \approx 5957 \text{ milj.}$ $2015: 5957 \cdot 0,07 = 416,99$ $5957 + 416,99 = 6373,99 \approx 6374 \text{ milj.}$ $2016: 6374 \cdot 0,07 = 446,18$ $6374 + 446,18 = 6820,18 \approx 6820 \text{ milj.}$ |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |  |  |  |   |  |   |  |   |  |  |   |   |  |  |  |   |  |   |  |
| Elevarbete 3                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0/2/1<br><table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td></td><td></td><td></td></tr><tr><td>B</td><td></td><td>X</td><td></td></tr><tr><td>M</td><td></td><td></td><td>/</td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>K</td><td></td><td>X</td><td></td></tr></table> |   | E | C | A | P |  |  |  | B |  | X |  | M |  |  | / | R |  |  |  | K |  | X |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | E                                                                                                                                                                                                                                                                                                                 | C | A |   |   |   |  |  |  |   |  |   |  |   |  |  |   |   |  |  |  |   |  |   |  |
| P                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |  |  |  |   |  |   |  |   |  |  |   |   |  |  |  |   |  |   |  |
| B                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                   | X |   |   |   |   |  |  |  |   |  |   |  |   |  |  |   |   |  |  |  |   |  |   |  |
| M                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                   |   | / |   |   |   |  |  |  |   |  |   |  |   |  |  |   |   |  |  |  |   |  |   |  |
| R                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |  |  |  |   |  |   |  |   |  |  |   |   |  |  |  |   |  |   |  |
| K                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                   | X |   |   |   |   |  |  |  |   |  |   |  |   |  |  |   |   |  |  |  |   |  |   |  |
| $4863 + 7\% \quad 4863 \cdot 1,07 = 5203,41 \quad (2012)$ $5203,41 \cdot 1,07 = 5567,6487 \quad (2013)$ $5567,6487 \cdot 1,07 = 5957,38411 \quad (2014)$ $5957,38411 \cdot 1,07 = 6374,401 \quad (2015)$ $6374,401 \cdot 1,07 = 6820,6 \quad (2016)$ <p>Svar: 6821 milj. kr.</p>                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |  |  |  |   |  |   |  |   |  |  |   |   |  |  |  |   |  |   |  |

|              |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |  |  |  |   |  |   |  |   |  |  |   |   |  |  |  |   |  |   |  |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|--|--|--|---|--|---|--|---|--|--|---|---|--|--|--|---|--|---|--|
| Elevarbete 4 | 0/2/2<br><table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td></td><td></td><td></td></tr><tr><td>B</td><td></td><td>X</td><td></td></tr><tr><td>M</td><td></td><td></td><td>X</td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>K</td><td></td><td>X</td><td></td></tr></table> |   | E | C | A | P |  |  |  | B |  | X |  | M |  |  | X | R |  |  |  | K |  | X |  |
|              | E                                                                                                                                                                                                                                                                                                                 | C | A |   |   |   |  |  |  |   |  |   |  |   |  |  |   |   |  |  |  |   |  |   |  |
| P            |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |  |  |  |   |  |   |  |   |  |  |   |   |  |  |  |   |  |   |  |
| B            |                                                                                                                                                                                                                                                                                                                   | X |   |   |   |   |  |  |  |   |  |   |  |   |  |  |   |   |  |  |  |   |  |   |  |
| M            |                                                                                                                                                                                                                                                                                                                   |   | X |   |   |   |  |  |  |   |  |   |  |   |  |  |   |   |  |  |  |   |  |   |  |
| R            |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |  |  |  |   |  |   |  |   |  |  |   |   |  |  |  |   |  |   |  |
| K            |                                                                                                                                                                                                                                                                                                                   | X |   |   |   |   |  |  |  |   |  |   |  |   |  |  |   |   |  |  |  |   |  |   |  |
| Elevarbete 5 | 0/2/2<br><table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td></td><td></td><td></td></tr><tr><td>B</td><td></td><td>X</td><td></td></tr><tr><td>M</td><td></td><td></td><td>X</td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>K</td><td></td><td>X</td><td></td></tr></table> |   | E | C | A | P |  |  |  | B |  | X |  | M |  |  | X | R |  |  |  | K |  | X |  |
|              | E                                                                                                                                                                                                                                                                                                                 | C | A |   |   |   |  |  |  |   |  |   |  |   |  |  |   |   |  |  |  |   |  |   |  |
| P            |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |  |  |  |   |  |   |  |   |  |  |   |   |  |  |  |   |  |   |  |
| B            |                                                                                                                                                                                                                                                                                                                   | X |   |   |   |   |  |  |  |   |  |   |  |   |  |  |   |   |  |  |  |   |  |   |  |
| M            |                                                                                                                                                                                                                                                                                                                   |   | X |   |   |   |  |  |  |   |  |   |  |   |  |  |   |   |  |  |  |   |  |   |  |
| R            |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |  |  |  |   |  |   |  |   |  |  |   |   |  |  |  |   |  |   |  |
| K            |                                                                                                                                                                                                                                                                                                                   | X |   |   |   |   |  |  |  |   |  |   |  |   |  |  |   |   |  |  |  |   |  |   |  |

|                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |  |   |  |   |  |   |  |  |  |   |  |  |  |   |  |  |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|--|---|--|---|--|---|--|--|--|---|--|--|--|---|--|--|--|
| Elevarbete                                                                                                                                                                                                                                                                                                                                                  | 1/0/0                                                                                                                                                                                                                                                                                                    |   |   |   |   |   |   |   |  |   |  |   |  |   |  |  |  |   |  |  |  |   |  |  |  |
| $200g = 25 \text{ kr}$<br>$25 \cdot 5 = 125$<br>kilopriset från början 125 kr/kg                                                                                                                                                                                                                                                                            | <table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td>X</td><td></td><td></td></tr><tr><td>B</td><td></td><td></td><td></td></tr><tr><td>M</td><td></td><td></td><td></td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>K</td><td></td><td></td><td></td></tr></table>   |   | E | C | A | P | X |   |  | B |  |   |  | M |  |  |  | R |  |  |  | K |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                             | E                                                                                                                                                                                                                                                                                                        | C | A |   |   |   |   |   |  |   |  |   |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P                                                                                                                                                                                                                                                                                                                                                           | X                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |  |   |  |   |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |  |   |  |   |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |  |   |  |   |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |  |   |  |   |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |  |   |  |   |  |   |  |  |  |   |  |  |  |   |  |  |  |
| Elevarbete 2                                                                                                                                                                                                                                                                                                                                                | 1/1/0                                                                                                                                                                                                                                                                                                    |   |   |   |   |   |   |   |  |   |  |   |  |   |  |  |  |   |  |  |  |   |  |  |  |
| $\frac{200}{100} = 2 \cdot 10 = 20 \quad 200 + 20 = 220$<br>$\frac{25}{10} = 2,5 \cdot 3 = 6 + 1,5 = 7,5 \quad 25 + 7,5 = 32,5 \text{ kr}$<br>$5 \cdot 25 = 125 \text{ kr/kg}$<br>$5 \cdot 32,5 = 162,5 \quad \text{Svar: } 30\%$                                                                                                                           | <table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td>X</td><td></td><td></td></tr><tr><td>B</td><td></td><td>X</td><td></td></tr><tr><td>M</td><td></td><td></td><td></td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>K</td><td></td><td></td><td></td></tr></table>  |   | E | C | A | P | X |   |  | B |  | X |  | M |  |  |  | R |  |  |  | K |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                             | E                                                                                                                                                                                                                                                                                                        | C | A |   |   |   |   |   |  |   |  |   |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P                                                                                                                                                                                                                                                                                                                                                           | X                                                                                                                                                                                                                                                                                                        |   |   |   |   |   |   |   |  |   |  |   |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                          | X |   |   |   |   |   |   |  |   |  |   |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |  |   |  |   |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |  |   |  |   |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |  |   |  |   |  |   |  |  |  |   |  |  |  |   |  |  |  |
| Elevarbete 3                                                                                                                                                                                                                                                                                                                                                | 1/2/0                                                                                                                                                                                                                                                                                                    |   |   |   |   |   |   |   |  |   |  |   |  |   |  |  |  |   |  |  |  |   |  |  |  |
| $\frac{25}{0,20} = 125 \text{ kr/kg}$<br>Priset ökar med 30%<br>$2,5 \cdot 3 = 7,5$<br>Nytt pris = 32,5 kr<br>Ny vikt = 220g<br>$\frac{32,5}{0,22} = 147,72 (\approx 150 \text{ kr/kg})$<br>$10\% \text{ av } 125 = 12,5$<br>$12,5 \cdot 2 = 25$<br>$125 + 25 = 150 \text{ kr}$<br>Priset ökar med 20%                                                      | <table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td>X</td><td>X</td><td></td></tr><tr><td>B</td><td></td><td>X</td><td></td></tr><tr><td>M</td><td></td><td></td><td></td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>K</td><td></td><td></td><td></td></tr></table> |   | E | C | A | P | X | X |  | B |  | X |  | M |  |  |  | R |  |  |  | K |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                             | E                                                                                                                                                                                                                                                                                                        | C | A |   |   |   |   |   |  |   |  |   |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P                                                                                                                                                                                                                                                                                                                                                           | X                                                                                                                                                                                                                                                                                                        | X |   |   |   |   |   |   |  |   |  |   |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                          | X |   |   |   |   |   |   |  |   |  |   |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |  |   |  |   |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |  |   |  |   |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |  |   |  |   |  |   |  |  |  |   |  |  |  |   |  |  |  |
| Elevarbete 4                                                                                                                                                                                                                                                                                                                                                | 1/2/0                                                                                                                                                                                                                                                                                                    |   |   |   |   |   |   |   |  |   |  |   |  |   |  |  |  |   |  |  |  |   |  |  |  |
| $200 \cdot 0,1 = 20 \quad \text{ny vikt: } 220g$<br>$25 \cdot 0,3 = 7,5 \quad 25 + 7,5 = \text{nytt pris} = 32,5 \text{ kr}$<br>$200x = 1000 \quad x = 5 \quad \frac{325}{220} = 0,15$<br>$5 \cdot 200 = 1000g$<br>$25 \cdot 5 = 125 \text{ kr/kg} \quad \text{nytt pris} \approx 150 \text{ kr/kg}$<br>$\frac{25}{150} \approx 0,17 = 17\% \text{ ökning}$ | <table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td>X</td><td>X</td><td></td></tr><tr><td>B</td><td></td><td>X</td><td></td></tr><tr><td>M</td><td></td><td></td><td></td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>K</td><td></td><td></td><td></td></tr></table> |   | E | C | A | P | X | X |  | B |  | X |  | M |  |  |  | R |  |  |  | K |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                             | E                                                                                                                                                                                                                                                                                                        | C | A |   |   |   |   |   |  |   |  |   |  |   |  |  |  |   |  |  |  |   |  |  |  |
| P                                                                                                                                                                                                                                                                                                                                                           | X                                                                                                                                                                                                                                                                                                        | X |   |   |   |   |   |   |  |   |  |   |  |   |  |  |  |   |  |  |  |   |  |  |  |
| B                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                          | X |   |   |   |   |   |   |  |   |  |   |  |   |  |  |  |   |  |  |  |   |  |  |  |
| M                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |  |   |  |   |  |   |  |  |  |   |  |  |  |   |  |  |  |
| R                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |  |   |  |   |  |   |  |  |  |   |  |  |  |   |  |  |  |
| K                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |  |   |  |   |  |   |  |  |  |   |  |  |  |   |  |  |  |
| Kommentar: Jämför med det nya priset, vilket inte är korrekt.                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |  |   |  |   |  |   |  |  |  |   |  |  |  |   |  |  |  |

Elevarbete 5

$$\frac{200g}{25kr}$$

$$200g = 25kr$$

$$1kg = 125kr$$

$$\frac{200g + 10\%}{25 + 30\%}$$

$$200 \cdot 0,1 = 20$$

$$200 + 20 = 220g$$

$$25 \cdot 0,3 = 7,5$$

$$25 + 7,5 = 32,5kr$$

$$\frac{220g}{11} = \frac{32,5}{11}$$

$$20g^{50} = 2,9545454^{50}$$

$$1kg = 147,7kr$$

$$147,7 - 125 = 22,7$$

$$\frac{22,7}{125} = 0,1816 \approx 18\%$$

Svar: Kilopriset ändrades med 18% (höjdes).

1/2/1

|   |   |   |   |
|---|---|---|---|
|   | E | C | A |
| P | X | X | X |
| B |   | X |   |
| M |   |   |   |
| R |   |   |   |
| K |   |   |   |

|                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                 |   |   |   |   |   |  |  |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|--|--|--|---|--|--|--|---|--|---|--|---|--|--|--|---|--|--|--|
| Elevarbete 1                                                                                                                                                                                                                                                                                                                                                                                               | 0/1/0<br><table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td></td><td></td><td></td></tr><tr><td>B</td><td></td><td></td><td></td></tr><tr><td>M</td><td></td><td>x</td><td></td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>K</td><td></td><td></td><td></td></tr></table> |   | E | C | A | P |  |  |  | B |  |  |  | M |  | x |  | R |  |  |  | K |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                            | E                                                                                                                                                                                                                                                                                                               | C | A |   |   |   |  |  |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| P                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                 |   |   |   |   |   |  |  |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| B                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                 |   |   |   |   |   |  |  |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| M                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                 | x |   |   |   |   |  |  |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| R                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                 |   |   |   |   |   |  |  |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| K                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                 |   |   |   |   |   |  |  |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| <p>b/ <math>50x + 120(92 - x) = 8380</math><br/><math>50x + 120 \cdot 92 - x = 8380</math><br/><math>49x + 120 \cdot 92 = 8380</math><br/><math>49x + 11040 = 8380</math><br/><math>49x + 11040 - 8380 = 8380 - 8380</math><br/><math>49x + 2660 =</math><br/><math>49x + 2660 - 2660 = -2660</math><br/><math display="block">\frac{49x}{49} = \frac{-2660}{49}</math><br/><math>x \approx -54</math></p> |                                                                                                                                                                                                                                                                                                                 |   |   |   |   |   |  |  |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| Kommentar: Korrekt användning av likhetstecknet.                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                 |   |   |   |   |   |  |  |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |

|                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                 |   |   |   |   |   |  |   |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|--|---|--|---|--|--|--|---|--|---|--|---|--|--|--|---|--|--|--|
| Elevarbete 2                                                                                                                                                                                         | 0/1/0<br><table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td></td><td></td><td></td></tr><tr><td>B</td><td></td><td></td><td></td></tr><tr><td>M</td><td></td><td>x</td><td></td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>K</td><td></td><td></td><td></td></tr></table> |   | E | C | A | P |  |   |  | B |  |  |  | M |  | x |  | R |  |  |  | K |  |  |  |
|                                                                                                                                                                                                      | E                                                                                                                                                                                                                                                                                                               | C | A |   |   |   |  |   |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| P                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                 |   |   |   |   |   |  |   |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| B                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                 |   |   |   |   |   |  |   |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| M                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                 | x |   |   |   |   |  |   |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| R                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                 |   |   |   |   |   |  |   |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| K                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                 |   |   |   |   |   |  |   |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| <p>b/ <math>50x + 120(92 - x) = 8380</math><br/><math>50 \cdot 38 \neq 1900 \quad 120 \cdot (92 - 38) = 6480</math><br/><math>1900 + 6480 = 8380</math></p> <p>c/ barn = 38 st<br/>vuxna = 54 st</p> | 0/1/0<br><table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td></td><td>x</td><td></td></tr><tr><td>B</td><td></td><td></td><td></td></tr><tr><td>M</td><td></td><td></td><td></td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>K</td><td></td><td></td><td></td></tr></table> |   | E | C | A | P |  | x |  | B |  |  |  | M |  |   |  | R |  |  |  | K |  |  |  |
|                                                                                                                                                                                                      | E                                                                                                                                                                                                                                                                                                               | C | A |   |   |   |  |   |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| P                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                 | x |   |   |   |   |  |   |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| B                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                 |   |   |   |   |   |  |   |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| M                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                 |   |   |   |   |   |  |   |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| R                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                 |   |   |   |   |   |  |   |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| K                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                 |   |   |   |   |   |  |   |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| Kommentar: Ekvationslösningen är inte redovisad. Troligtvis löst med prövning.                                                                                                                       |                                                                                                                                                                                                                                                                                                                 |   |   |   |   |   |  |   |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |

|                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |  |   |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |   |   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|--|---|--|---|--|--|--|---|--|---|--|---|--|--|--|---|--|---|---|
| Elevarbete 3                                                                                                                                                                                                                                                                                                                                                                       | 0/2/0<br><table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td></td><td></td><td></td></tr><tr><td>B</td><td></td><td></td><td></td></tr><tr><td>M</td><td></td><td>x</td><td></td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>K</td><td></td><td></td><td></td></tr></table>   |   | E | C | A | P |  |   |  | B |  |  |  | M |  | x |  | R |  |  |  | K |  |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                    | E                                                                                                                                                                                                                                                                                                                 | C | A |   |   |   |  |   |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |   |   |
| P                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |  |   |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |   |   |
| B                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |  |   |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |   |   |
| M                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                   | x |   |   |   |   |  |   |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |   |   |
| R                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |  |   |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |   |   |
| K                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |  |   |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |   |   |
| <p>b/ <math>5x + 120(92 - x) = 8380</math><br/><math>5x + 11040 - 120x = 8380</math><br/><math>5x + 11040 - 8380 = 120x</math><br/><math>5x - 5x + 2660 = 120x - 5x</math><br/><math display="block">\frac{2660}{115} = \frac{115x}{115}</math><br/><math>23,13 \approx x</math></p> <p>c/ <math>x \approx 23</math> barn<br/><math>92 - 23,13 = 68,87 \approx 69</math> vuxna</p> | 0/2/1<br><table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td></td><td>x</td><td></td></tr><tr><td>B</td><td></td><td></td><td></td></tr><tr><td>M</td><td></td><td></td><td></td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>K</td><td></td><td>x</td><td>x</td></tr></table> |   | E | C | A | P |  | x |  | B |  |  |  | M |  |   |  | R |  |  |  | K |  | x | x |
|                                                                                                                                                                                                                                                                                                                                                                                    | E                                                                                                                                                                                                                                                                                                                 | C | A |   |   |   |  |   |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |   |   |
| P                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                   | x |   |   |   |   |  |   |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |   |   |
| B                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |  |   |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |   |   |
| M                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |  |   |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |   |   |
| R                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |  |   |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |   |   |
| K                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                   | x | x |   |   |   |  |   |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |   |   |
| Kommentar: Ekvationen felaktigt avskriven och ger därför ett orimligt svar med decimaler. Eleven tolkar dock detta till rimligt antal barn och vuxna.                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |  |   |  |   |  |  |  |   |  |   |  |   |  |  |  |   |  |   |   |

Elevarbete 4

b)  $50x + 120(92 - x) = 8380$   
 $50x + 11040 - 120x = 8380$   
 $11040 - 8380 = 120x - 50x$   
 $2660 = 70x$   
 $38 = x$

c) 38 barn  
 54 vuxna

Kommentar: Eleven löser ekvationen korrekt men ger ingen förklaring till antalet vuxna.

0/2/1

|   |   |   |   |
|---|---|---|---|
|   | E | C | A |
| P |   |   |   |
| B |   |   |   |
| M | x | x | x |
| R |   |   |   |
| K |   |   |   |

0/1/0

|   |   |   |   |
|---|---|---|---|
|   | E | C | A |
| P |   | x |   |
| B |   |   |   |
| M |   |   |   |
| R |   |   |   |
| K |   |   |   |

Elevarbete 5

a) barn

b)  $50x + 120(92 - x) = 8380$   
 $50x + 11040 - 120x = 8380$   
 $11040 - 8380 = 120x - 50x$   
 $2660 = 70x$   
 $38 = x$

c)  $50 \text{ kr} \cdot 38 = 1900 \text{ kr}$   
 $8380 - 1900 = 6480 \text{ kr}$   
 Antal vuxna:  $\frac{6480}{120} = 54$   
 Svar: 54 vuxna och 38 barn

Kommentar: Tydlig redovisning av hela problemet (a–c) med korrekt matematiskt språk (+A<sub>K</sub>).

1/4/2

|   |   |   |   |
|---|---|---|---|
|   | E | C | A |
| P | x | x |   |
| B |   |   |   |
| M | x | x | x |
| R |   |   |   |
| K | x | x |   |

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |   |   |  |   |   |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|--|---|---|---|--|--|--|---|--|---|--|---|--|--|--|---|--|--|--|
| Elevarbete 1 | <div>0/0/0</div> <table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td></td><td></td><td></td></tr><tr><td>B</td><td></td><td></td><td></td></tr><tr><td>M</td><td></td><td></td><td></td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>K</td><td></td><td></td><td></td></tr></table> <div>Ami - 26 ggr 2180 kr<br/>Liam - 34 ggr 2820 kr<br/>Ami <math>\frac{2180}{26} \approx 83,4</math> Liam <math>\frac{2820}{34} \approx 82,9</math><br/>De betalade ungefär 83 kr/gång</div>                                                                                                                        |   | E | C | A | P |  |   |   | B |  |  |  | M |  |   |  | R |  |  |  | K |  |  |  |
|              | E                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | C | A |   |   |   |  |   |   |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| P            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |   |   |  |   |   |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| B            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |   |   |  |   |   |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| M            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |   |   |  |   |   |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| R            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |   |   |  |   |   |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| K            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |   |   |  |   |   |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| Elevarbete 2 | <div>0/1/0</div> <table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td></td><td>X</td><td></td></tr><tr><td>B</td><td></td><td></td><td></td></tr><tr><td>M</td><td></td><td></td><td></td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>K</td><td></td><td></td><td></td></tr></table> <div>A = 2180 - 200 = 1980<br/>L = 2820 - 200 = 2620<br/><math>\frac{1980}{26} \approx 76,1</math>      <math>\frac{2620}{34} \approx 77</math><br/>Svar: Kortet kostar 200 kr och de betalade ungefär 76 kr per gång.<br/><br/>Kommentar: Förstår betalningssystemet men "gissar" fel pris på medlemskortet.</div> |   | E | C | A | P |  | X |   | B |  |  |  | M |  |   |  | R |  |  |  | K |  |  |  |
|              | E                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | C | A |   |   |   |  |   |   |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| P            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | X |   |   |   |   |  |   |   |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| B            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |   |   |  |   |   |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| M            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |   |   |  |   |   |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| R            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |   |   |  |   |   |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| K            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |   |   |  |   |   |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| Elevarbete 3 | <div>0/2/0</div> <table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td></td><td>X</td><td></td></tr><tr><td>B</td><td></td><td></td><td></td></tr><tr><td>M</td><td></td><td>X</td><td></td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>K</td><td></td><td></td><td></td></tr></table> <div>2820 - 2180 = 34x - 26x<br/>640 = 8x<br/>80 = x<br/>Svar: 80 kr i avgift</div>                                                                                                                                                                                                                                |   | E | C | A | P |  | X |   | B |  |  |  | M |  | X |  | R |  |  |  | K |  |  |  |
|              | E                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | C | A |   |   |   |  |   |   |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| P            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | X |   |   |   |   |  |   |   |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| B            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |   |   |  |   |   |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| M            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | X |   |   |   |   |  |   |   |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| R            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |   |   |  |   |   |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| K            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |   |   |  |   |   |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| Elevarbete 4 | <div>0/2/0</div> <table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td></td><td>X</td><td></td></tr><tr><td>B</td><td></td><td></td><td></td></tr><tr><td>M</td><td></td><td>X</td><td></td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>K</td><td></td><td></td><td></td></tr></table> <div>Ami: 26x + y = 2180<br/>Liam: 34x + y = 2820<br/>x = pris per gång<br/>y = medlemskort</div>                                                                                                                                                                                                                  |   | E | C | A | P |  | X |   | B |  |  |  | M |  | X |  | R |  |  |  | K |  |  |  |
|              | E                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | C | A |   |   |   |  |   |   |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| P            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | X |   |   |   |   |  |   |   |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| B            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |   |   |  |   |   |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| M            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | X |   |   |   |   |  |   |   |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| R            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |   |   |  |   |   |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| K            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |   |   |  |   |   |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| Elevarbete 5 | <div>0/2/1</div> <table><tr><td></td><td>E</td><td>C</td><td>A</td></tr><tr><td>P</td><td></td><td>X</td><td>X</td></tr><tr><td>B</td><td></td><td></td><td></td></tr><tr><td>M</td><td></td><td>X</td><td></td></tr><tr><td>R</td><td></td><td></td><td></td></tr><tr><td>K</td><td></td><td></td><td></td></tr></table> <div>x = medlemskort      y = kr per gång<br/>Ami <math>80y \cdot 26 + x^{100} = 2180</math><br/>Liam <math>80y \cdot 34 + x^{100} = 2820</math><br/>Svar: 80 kr och 100 kr<br/><br/>Kommentar: Uppställt ekvationssystem löses med prövning.</div>                                                               |   | E | C | A | P |  | X | X | B |  |  |  | M |  | X |  | R |  |  |  | K |  |  |  |
|              | E                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | C | A |   |   |   |  |   |   |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| P            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | X | X |   |   |   |  |   |   |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| B            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |   |   |  |   |   |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| M            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | X |   |   |   |   |  |   |   |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| R            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |   |   |  |   |   |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |
| K            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |   |   |  |   |   |   |  |  |  |   |  |   |  |   |  |  |  |   |  |  |  |



Elevarbete 6

$$2820 - 2180 = 640 \text{ skillnaden mellan deras kostnader}$$

$$34 - 26 = 8 \text{ skillnaden i gånger}$$

$$\frac{640}{8} = 80 \text{ kostnad per gång}$$

$$34 \cdot 80 = 2720$$

$$2820 - 2720 = 100$$

totala kostnaden minus  
kostnad för alla gånger

medlemskortet

Svar: 80 kr och 100 kr

$$\text{Kontrollräkning: } 34 \cdot 80 + 100 = 2820 \text{ kr}$$

$$26 \cdot 80 + 100 = 2180 \text{ kr}$$

Kommentar: Eleven använder en generell aritmetisk metod.

0/2/3

|   | E | C | A |
|---|---|---|---|
| P |   | x | x |
| B |   |   |   |
| M |   | x | x |
| R |   |   |   |
| K |   |   | x |

Elevarbete 7

$x$ : medlemskort  $y$ : antal gånger

$$x + 34y = 2820$$

$$x + 26y = 2180$$

$$(x + 34y) - (x + 26y) = 2820 - 2180$$

$$\frac{8y}{8} = \frac{640}{8}$$

$$y = 80$$

$$x + 34 \cdot 80 = 2820$$

$$x + 2720 = 2820$$

$$x = 2820 - 2720$$

$$x = 100$$

$$x + 26 \cdot 80 = 2180$$

$$x + 2080 = 2180$$

$$x = 2180 - 2080$$

$$x = 100$$

Svar: 80 kr och 100 kr

0/2/3

|   | E | C | A |
|---|---|---|---|
| P |   | x | x |
| B |   |   |   |
| M |   | x | x |
| R |   |   |   |
| K |   |   | x |